



Multichannel Motion Artefact Correction in fNIRS

Harry Burns

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Abstract

Functional near-infrared spectroscopy (fNIRS) is a non-invasive neuroimaging technique for measuring task-related haemodynamic change. Its practical usefulness is limited by sensitivity to motion artefacts (MAs), a category of structured noise, which can introduce disturbances that obscure genuine physiological responses. This project investigates motion artefact correction for high-density (HD) fNIRS data acquired using the Gowerlabs LUMO headset, a modern system with denser channel coverage than that used in much of the existing literature.

The main technical contribution is a cluster-based motion correction method combining **Global Variance of the Temporal Derivative (GVTD)** [1] with **targeted Principal Component Analysis (tPCA)** [2]. Channels are grouped into local source-detector clusters, GVTD is used to identify motion-contaminated periods within each cluster, and tPCA is then applied as a targeted correction step. To support this, a custom task-specific Python pipeline was developed to process SNIRF files (the standard fNIRS recording file format), reimplement published correction methods in a transparent way, and run preprocessing and evaluation for the experimental recordings.

The proposed method was developed and compared with baseline approaches including no correction and **Temporal Derivative Distribution Repair (TDDR)** [3] on finger-tapping recordings collected under different motion conditions. Although limitations in the quality and quantity of the collected data prevented reliable conclusions about comparative performance, the project provides a full implementation of the proposed method, the compared method, and a reproducible basis for further investigation of multichannel correction in dense fNIRS systems.

1 Introduction

Functional near-infrared spectroscopy (fNIRS) is a rapidly expanding non-invasive brain-imaging technique that uses near-infrared light to monitor haemodynamic changes in the outer layers of the brain. When neurons in the brain are active, their metabolic demand increases and so, to ensure they are not deprived of oxygen or glucose, local blood vessels will start to dilate in response to the

increased activity. Due to the difference in absorption spectra of oxygenated and deoxygenated blood (oxyhaemoglobin (HbO) and deoxyhaemoglobin (HbR)), information about the dilation of these blood vessels can be inferred through a change in the optical density received in the source/detector pairs. These measurements of localised changes provide a proxy measurement of brain activity in their corresponding regions.

fNIRS as a brain imaging modality has grown in popularity due to it being highly-portable (especially compared to other modalities such as fMRI), non-invasive, lower cost, and the fact that it can be used during physical activity, a quality not common among brain imaging modalities. These advantages come with the practical drawback of fNIRS signals often being contaminated by physiological noise, motion artefacts, and setup-related noise, all of which can distort the measured signal and make interpretation more difficult.

In fNIRS, optodes are placed on the scalp of the participant, making it less sensitive to motion artefacts arising from head and scalp movement than other brain imaging modalities [3]. Despite this, motion artefacts are still an extremely prevalent problem in fNIRS literature. Researchers have reported that the artefacts caused by subjects' movements can significantly influence the quality of fNIRS signals [4] by reducing the signal-to-noise ratio (SNR) [5]. Researchers have also shown that using MA correction techniques can reduce this negative effect, increasing classification accuracy in cognition experiments. [6]

Many different approaches to motion artefact correction have been proposed for fNIRS. Huang et al. [6] provides a broad review of techniques developed up to that point, outlining their origins, operating principles, implementation considerations, and underlying assumptions. A number of further studies have also compared the performance of these methods under different experimental settings. However, despite the variety of available approaches, the literature remains difficult to reproduce consistently in practice, as important implementation details and processing choices are often under-specified. This makes it difficult both to evaluate methods fairly and to assess how well they generalise to newer fNIRS systems. For this reason, part of this project involved implementing the relevant processing and correction methods directly in Python, such that each stage of the pipeline could be made explicit and controlled.

This project therefore investigates a cluster-based motion correction method for high-density fNIRS data. The main technical contribution is a new MA correction approach that combines Global Variance of the Temporal Derivative (GVTD) [1] with targeted Principal Component Analysis (tPCA) [2]. Channels are grouped into local clusters using a spatially motivated clustering rule based on source-detector locations and separations. GVTD is then used to identify motion-contaminated periods within each cluster, after which tPCA is applied as a targeted correction step.

To support this investigation, a custom task-specific Python pipeline was developed to process SNIRF recordings, reimplement published correction methods transparently, and run preprocessing and evaluation on experimental data. Baseline approaches including no correction and Temporal Derivative Distribution Repair (TDDR) were also implemented for comparison. TDDR was chosen as the main baseline because, like GVTD, it operates in the temporal-derivative domain, but unlike GVTD it is applied independently to each channel. This made it a useful point of comparison between single-channel and multichannel correction strategies.

Overall, the main outcome of this project is the implementation of a cluster-based GVTD-gated tPCA correction method together with a Python pipeline for loading SNIRF recordings, preprocessing signals, applying motion correction, and evaluating outputs.

Experimental recordings were collected under finger-tapping conditions with varying levels of imposed head motion, with the aim of assessing whether correction could recover more plausible task-related responses and reduce the impact of motion contamination.

Although limitations in the quality and quantity of the collected data restrict the strength of the final empirical conclusions, the project provides a full implementation of the proposed method and a structured experimental pipeline for investigating multichannel correction in dense fNIRS systems. The remainder of this report reviews the relevant literature, describes the proposed method and implementation, and discusses the extent to which the resulting experiments support its use.

2 Background and Related Work

2.1 Motion Artefacts in fNIRS

Motion artefacts in fNIRS arise when subject movement alters optode–scalp coupling, producing signal changes unrelated to the underlying haemodynamic response [2]. In higher-density fNIRS systems, these effects are often not isolated to a single measurement, but can affect multiple nearby channels at the same time [1].

In the recorded time series, motion artefacts are most commonly described as *spikes* and *baseline shifts*. Spikes appear as brief and abrupt fluctuations, while baseline shifts appear as step-like changes that displace the signal to a new level. Figure 1 shows examples of these characteristic shapes. [1, 3]

Some artefacts also appear in a slower drift-like form, which can be more difficult to distinguish from genuine haemodynamic trends. These slower distortions are not the primary focus of the correction methods considered in this project.

2.2 Existing Approaches to Motion Artefact Correction

A wide range of motion artefact correction methods have been proposed for fNIRS. Huang et al. [6] group these into hardware-based solutions, which use additional auxiliary measurements (for example an accelerometer), and signal-processing-based solutions, which

act directly from the recorded signals. In this project only signal-processing-based solutions are considered, as the aim is to develop an algorithm that operates directly on the recorded signal time series.

Within the signal-processing-based literature, methods mainly differ in: (i) their assumptions about artefact structure; (ii) whether they operate on channels independently (single-channel) or exploit relationships across multiple channels (multichannel); (iii) whether they require a motion detection step as a pre-requisite, and (iv) in which step of the preprocessing pipeline they are used. The vast majority of proposed methods are single-channel as it was only recently that high-density fNIRS systems were standardised, with the first commercially available HD system being the Gowerlabs LUMO released in 2019 [7]. Until then high-density systems were typically custom-built lab prototypes. Additionally, the majority of correction methods act upon the Optical Density (OD) signal, rather than the raw light intensity signal, and some even work directly on the HbO and HbR concentration signals themselves.

In the following subsections I will summarise the most common MA correction methods used, with Huang et al. [6] used as a main reference for their assumptions, implementation characteristics, and reported strengths and weaknesses.

2.2.1 Correlation-based Signal Improvement (CBSI)

Correlation-based signal improvement (CBSI) [8], proposed in 2010 is a correction method based on the three fundamental assumptions: (i) HbO and HbR concentrations should be strictly negatively correlated, (ii) motion artefacts affect both HbO and HbR identically, and (iii) motion artefacts are not correlated to the actual concentration changes. CBSI corrects the HbO concentrations by subtracting a scaled HbR signal to match the variance of the HbO. If all assumptions are met, this results in any positive correlation (motion artefacts) being removed. The HbR is then re-calculated by multiplying the HbO with the negative inverse of the scaling factor used beforehand.

CBSI acts directly on the HbO/HbR, meaning any preprocessing beforehand has a direct effect on the algorithms effectiveness. The assumptions made for CBSI, specifically assumption one, have been found to not always hold in experimental conditions, since HbO and HbR signals are not always strongly negatively correlated in real data [9], and so it is typically used as means of comparison of other methods.

2.2.2 Spline Interpolation / MARA

Spline interpolation, commonly referred to as the motion artefact reduction algorithm (MARA), is a single-channel MA correction method proposed by Scholkmann et al. in 2010 [10]. The method is based on two main assumptions: (i) the measured fNIRS signal can be represented as a linear sum of motion artefacts and the motion-free signal, and (ii) within motion-corrupted segments, the artefact component dominates the measured signal. MARA therefore operates in two stages: first detecting and segmenting motion-contaminated regions, and then reconstructing those regions using spline interpolation.

MARA identifies candidate artefact segments using a moving standard deviation threshold, after which spline fitting is used to estimate and remove the corrupted component from each segment. It can be applied to optical intensities (raw signal), OD, or HbO/HbR. It has shown good performance in some comparative studies, although later work suggests that its effectiveness is strongly dependent on accurate artefact segmentation and may vary across datasets [10, 9, 6].

2.2.3 Wavelet-Based Methods

Wavelet-based MA correction, proposed by Molavi and Dumont in 2012 [11], is a single-channel correction method that assumes the measured fNIRS signal can be represented as the sum of a motion-free signal and an artefact component. The method transforms the recorded signal into the discrete wavelet domain, where motion artefacts are assumed to appear as abnormal wavelet coefficients. These coefficients are then identified and removed or thresholded before the signal is reconstructed.

Like spline interpolation, wavelet-based correction can be applied to optical intensities, OD, or HbO/HbR. It has been reported to perform well in comparative studies, particularly for spike-like artefacts, although its effectiveness depends strongly on the thresholding strategy and chosen parameters [6] as, unlike MARA, it does not come with its own motion detection step.

2.3 Selected Approaches

2.3.1 tPCA

One of the earlier MA correction approaches introduced was Principal Component Analysis (PCA) [12]. It is a fundamentally multi-channel approach that uses shared spatial covariation among channels to identify and remove the MA component of the signal across all channels at once. Targeted Principal Component Analysis (tPCA) is a variant of PCA that performs it only on motion-detected periods of time rather than the entire signal. The two main assumptions it uses are as follows: (i) motion artefacts appear in multiple channels with a common temporal variation; and (ii) motion artefacts account for the majority of the variance in the signal [2]. It is typically applied to the OD signal, but variations exist that are applied to other domains [6].

tPCA is a gated motion artefact correction approach, meaning it requires a motion detection step beforehand to function. This motion detection step is not native to tPCA and is instead to be selected by whoever is implementing it. In most tPCA comparisons

and implementations, motion-contaminated epochs are identified using the Homer/Homer2 (a standard library for fNIRS processing in MATLAB) thresholding approach, in which artefacts are detected on a channel-by-channel basis when either the absolute amplitude change or the change relative to the channel standard deviation exceeds a predefined threshold within a short time window. If an artefact is detected in any single channel, a bin spanning a short interval is placed around the event, and the bin is treated as a contaminated time-span across all channels.

Once all the periods of motion artefact contamination have been identified, they are concatenated temporally to create a signal $X \in \mathbb{R}^{T \times N}$, where T is the total number of samples spanned by all concatenated bins, and N is the number of channels being operated upon. PCA is then applied to X to decompose the shared variation across channels into orthogonal principal components, ordered by the proportion of variance they explain. The leading components are assumed to represent the dominant motion-related structure within the contaminated epochs and are projected out, after which the corrected bins are reinserted into their original temporal positions in the full signal [2]. Typically, the number of principal components to remove is selected to account for 97% of the variance, as is stated in its introductory paper. This process, including the motion detection step, is repeated twice in the original paper as an additional correction pass.

The fNIRS tPCA literature does not appear to specify a standard minimum number of channels. Existing reviews note that tPCA's performance depends on having multiple channels and can degrade when the number of channels is limited [9]. Because of this, the minimum channel threshold (8 channels) used in this work was a design choice rather than an fNIRS standard.

A weakness of tPCA is that the channels included in the correction step should be expected to share similar motion-related structure. If channels with very different measurement characteristics are combined, the leading components may reflect a mixture of effects that is not well matched to every channel, making the resulting correction less effective [2]. Because of this, it is sensible to group together channels with similar source-detector separations when applying the method.

tPCA has been shown to outperform wavelet-based and spline-based approaches in some different fNIRS setups [6], but its effectiveness is highly dependent on the performance of the motion detection step performed before [2]. Despite this, motion detection steps are typically single-channel in existing literature and not commonly experimented upon. This acted as a large part of the motivation of this project, since it leaves open the question of whether multichannel correction can be improved by replacing the single-channel detection stage with a multichannel alternative, especially given a high-density fNIRS system.

2.3.2 GVTD

Rather than being a correction algorithm like the aforementioned, the Global Variance of the Temporal Derivative (GVTD) (Sherafati et al. [1], 2020) is a motion detection algorithm that, when combined with a threshold cut-off value, creates a channel-wide motion-identifying mask. In its introductory paper, GVTD was shown to have "strong correlation with external measures of motion" and "high sensitivity and specificity to instructed motion" [1].

In practice, GVTD is often computed after temporal filtering, so that slow baseline drift and higher-frequency noise are less likely to dominate the signal's temporal derivative. Sherafati et al. [1] evaluated GVTD at multiple stages of the processing pipeline and found that it performed best after temporal filtering.

The GVTD formula is as follows

$$\mathbf{g} = [g_1, \dots, g_T], \quad g_i = \sqrt{\frac{1}{N} \sum_{j=1}^N (TD_{j,i})^2}, \quad TD_{j,i} = (y_{j,i} - y_{j,i-1}), \quad g_i \in \mathbb{R}_{>0}$$

where $\mathbf{g}$ is the GVTD vector, $y_{j,i} \in \mathbb{R}$ is the OD in channel j at time point i , and N, T are the number of channels and time samples [1].

The noise threshold (g_{thresh}) is based on the GVTD distribution mode ($\tilde{\kappa}$) plus a constant (c) times the standard deviation computed on the left (low) side of the mode (σ_L).

$$g_{\text{thresh}} = \tilde{\kappa} + c\sigma_L \quad \text{where} \quad \sigma_L = \sqrt{\frac{1}{n_L} \sum_{g_i < \tilde{\kappa}} (g_i - \tilde{\kappa})^2}, \quad n_L = \|\{g_i < \tilde{\kappa}, \forall g_i \in \mathbf{g}\}\|$$

One issue of using GVTD as a motion mask is the free parameter c that must be set for each recording. Sherafati et al. [1] reported that c varies greatly, recommending 10 for resting-state analysis and 3 for task-based analysis. The variable c represents the noise/signal trade off, where a lower c is more aggressive in motion detection.

In addition to the algorithm, the Sherafati et al. [1] introduces a channel filter based on the GVTD that works as follows: "Noisy measurements were empirically defined as those with greater than 7.5% temporal standard deviation in the least noisy (lowest mean GVTD) 60 seconds of each run, and were excluded from further processing".

GVTD has been mentioned and used in many different papers [1, 6, 13, 14], but mostly only for trial rejection and censoring. This, combined with the lack of a multichannel filter for tPCA, provided great motivation for evaluating their effectiveness when combined. To the best of my knowledge, GVTD and tPCA have not been explicitly combined in prior work.

2.3.3 TDDR

Temporal Derivative Distribution Repair (TDDR), proposed by Fishburn et al. in 2019 [3], is a single-channel MA correction method that operates directly on the temporal derivative of the signal. Unlike methods such as MARA and tPCA, it does not require a separate motion detection stage or any user-chosen detection thresholds. Instead, it assumes that: (i) non-motion fluctuations in the derivative are approximately normally distributed, (ii) most fluctuations contain no motion artefact contribution, and (iii) motion artefacts, when present, produce derivatives of much greater magnitude than normal fluctuations. Under these assumptions, motion-related fluctuations appear as large outliers in the derivative distribution and can be down-weighted using robust regression. [3]

The algorithm, performed individually for each channel, computes the temporal derivative (TD), initializes observation weights (1), and then iteratively estimates a weighted mean (2), residuals (3), and a robust scale parameter (4) before updating the weights using Tukey's biweight function (5), until the mean converges, at which point the TD is transformed by the final weights (6).

$$\begin{aligned}
 (1) \quad w_i &= 1 & (2) \quad \mu &= \frac{1}{\sum_i w_i} \sum_i w_i TD_i & (3) \quad r_i &= |TD_i - \mu| & (4) \quad \sigma &= 1.4826 * \text{median}(r_i) \\
 (5) \quad d_i &= \frac{r_i}{4.685 \sigma}, \quad w_i = \begin{cases} (1 - d_i^2)^2, & d_i < 1, \\ 0, & \text{otherwise} \end{cases} & (6) \quad \widetilde{TD}_i &= w_i (TD_i - \mu)
 \end{aligned}$$

The corrected derivative is then integrated to reconstruct the corrected signal. Because it works in the temporal-derivative domain, TDDR is particularly suited to abrupt artefacts such as spikes and baseline shifts. [3]

A major motivation for TDDR was to provide a parameter-free alternative to earlier methods. In the original paper, wavelet-based methods were noted to handle motion spikes well but to exacerbate baseline shifts, while MARA and tPCA were criticised for relying on user-supplied motion-detection parameters. TDDR was introduced as a method that removes both spikes and baseline shifts without this parameter-selection burden, and in the original validation it achieved the strongest activation-detection performance among the compared methods. [3]

One limitation of TDDR is that high-frequency components inflate the variance of the temporal derivative, similarly to GVTD, and reduce the effectiveness of the robust weighting stage. For this reason, the original implementation applies TDDR to a low-frequency component of the signal when the sampling rate is high, before adding back the uncorrected high-frequency residual. This means TDDR is most effective for large, abrupt artefacts, while smaller or slower artefacts may remain.

TDDR was chosen as the main baseline for three reasons. First, it has been reported to perform strongly on abrupt artefacts such as spikes and baseline shifts, which are also the main artefact types targeted by GVTD and tPCA. Second, like GVTD, it operates in the temporal-derivative domain, making it a useful comparison point between single-channel and multichannel derivative-based approaches. Third, it is widely regarded in recent fNIRS literature as an effective MA correction method.

2.4 Summary

Broadly, comparative studies reviewed by Huang et al. [6] suggest that no single correction method is universally best. Various methods have shown good performance under different conditions, with results depending on the dataset, artefact type, and evaluation metric used.

3 Data and Experimental Setup

As part of the evaluation of my proposed method, and as a means of empirically evaluating my processing pipeline, a part of this project was dedicated to data collection. The purpose of the recordings is to produce a predictable motor-task response under varying degrees of instructed motion, ranging from no motion to substantial head motion, in order to assess the efficacy of my proposed method.

In the fNIRS motion-correction literature, evaluation has been carried out using experimental, simulated, and semi-simulated data. In semi-simulated approaches, desired responses may be added to motion-artefact-corrupted resting data, or resting experimental data may be combined with desired responses and a designed artefact sequence. Using simulated data means you can model thousands of hours worth of recordings with whatever protocol you want, allowing for deep-learning-based approaches [15], but comes with a risk of model bias. When considering multichannel LUMO-like setups with channels placed in specific locations, the modelling process dramatically increases in complexity and would be its own entire project to explore. For this reason, simulated data was not used in this project. Similarly, no suitable pre-existing datasets were identified, as publicly available fNIRS recordings were not acquired using the LUMO headset, which is the system available at the university and whose channel density and spatial layout the proposed method is designed around. Instead, a set of instructed recordings were taken.

Recordings were collected from 5 participants using the Gowerlabs LUMO headset under instructed conditions. The following sections describe the hardware, recording protocol, the dataset collected, and considerations after a review of the dataset.

Unfortunately the recordings taken were all of very low quality. This has had a large impact on the entire evaluation portion of this project, removing the ability to use standard evaluation techniques. This is discussed in more detail at the end of this section.

3.1 Gowerlabs LUMO System

The hardware used for both data collection and method development was the Gowerlabs LUMO headset, a high-density fNIRS system released in 2019. The headset consists of a flexible cap populated by hexagonal nodes. In the recordings used for this project, 12 nodes were available in total, although only 6 were positioned sufficiently close to the target brain region to be considered for analysis.

Each node contains 3 light sources and 4 detectors, giving 36 sources and 48 detectors across the full headset. Measurements were acquired at two wavelengths, 735 nm and 850 nm. In the raw recording representation, each source-detector pair is associated with one measurement at each wavelength, giving a theoretical total of $36 \times 48 \times 2 = 3456$ source-detector-wavelength combinations. Each recorded time series was sampled at 6.25 Hz, so a recording of length t seconds may be represented as a signal array with dimensions $3456 \times \lfloor 6.25t \rfloor$.

The nodal structure of the LUMO headset provides an excellent basis for experimenting with ideas of spatial locality. Because measurements arise from an explicitly node-based arrangement of sources and detectors, the system naturally admits multiple ways of defining individual clusters of channels. This creates scope to investigate whether motion artefacts can be identified or corrected more effectively by exploiting local spatial relationships, rather than treating channels as entirely independent or processing the full set globally.

3.2 Recording Protocol

3.2.1 Task Selection

While not all fNIRS motion artefact comparison papers use a task-based evaluation, some instead choosing to look at the impact of MAs on resting-state data, task-based data is often easier to invoke and monitor the conditions of, and so it was suitable for this evaluation.

The role of the task is to provide a predictable and reproducible response in an area of the brain that can be easily captured using fNIRS hardware. Given fNIRS uses light intensity as its signal basis, one would expect that deeper areas of the brain elicit a lower signal-to-noise ratio due to more light being scattered while traversing the head [16]. This means an ideal response to monitor is nearby the scalp.

The tasks most commonly used in fNIRS studies broadly fall into motor and cognitive categories. For this project, a motor task, right hand finger tapping, was selected because it evokes a predictable and reproducible response in a single superficial cortical region that is well suited to fNIRS measurement, while also being straightforward to explain, perform, and monitor. Additionally, compared to cognitive tasks, finger tapping is easier to perform consistently while varying the degree of instructed head motion.

3.2.2 Motion Conditions

Each recording included three identical trials with varied instructed motion. These three categories, inspired by Al-Omairi et al. [17], are No Head Motion (NHM), Small Head Motion (SHM) and Large Head Motion (LHM). During NHM periods there will be no instructed motion and during SHM and LHM there will be 1 second of instructed motion per 30 s task-rest block.

While SHM stands for Small Head Motion, in this project it is represented by facial movements like raising the eyebrows and moving the jaw. These movements were found to cause motion artefacts in the LUMO headset in Xia et al. [15]. LHM will be shaking of the head left and right rapidly, aiming to cause the headset to jolt. In this project, rather than just one motion condition, both SHM and LHM were used to add redundancy in case one of them did not induce proper artefacts, and to explore any relationship between motion type and correction ability.

The trials were ordered in cascading amounts of induced motion in order to ensure that the "cleaner" trials had the cleanest initial pairing, as it was likely that the LHM trial could cause source-detector decoupling.

3.2.3 Block Design

As mentioned, each recording consisted of three structurally identical trials (ignoring the instructed motion). A 20 s baseline period was placed before the first trial, between consecutive trials, and after the final trial, such that adjacent trials shared the intervening baseline period.

Each trial consisted of 8 30 s task-rest blocks, consisting of a 10 s task period and a 20 s rest period. Task and rest periods were indicated by visual cues, and corresponding event markers were recorded alongside the fNIRS data using the LUMOX software. These markers were later used to segment the recordings for preprocessing, motion analysis, and block-averaged evaluation. Countdowns were not used as part of the visual indication in order to avoid introducing a preparatory response before the true task onset. Advance warning cues can themselves alter cortical activation during the motor preparation stage, so removing a countdown helped keep the recorded response more closely tied to the task period itself rather than to anticipation of movement [18].

The use of 10 s task periods and 20 s rest periods was chosen to match the relatively slow haemodynamic response measured by fNIRS. In fNIRS, the response usually peaks a few seconds after task onset and does not return to baseline immediately, so a longer rest period helps reduce carry-over between consecutive task blocks. This choice is also consistent with prior motor-task fNIRS studies, including the NHM/SHM/LHM finger-tapping protocol used by Al-Omairi et al. [17], which used 10 s tapping periods followed by 20 s rest specifically to allow the haemodynamic response to settle back towards baseline between blocks. The additional 20 s baseline periods before the first trial, between trials, and after the final trial were included to provide stable resting segments and to ensure that each trial began from a comparable baseline state, and also to provide optional trial-specific baseline periods to use in processing.

As mentioned, there was 1 s of instructed motion in each 30 s block. This was kept brief so that motion artefacts remained relatively uncommon within the recording, in line with the assumptions underlying GVTD and TDDR [3, 1].

All baseline, task, rest and motion onset and end times were marked with their corresponding time-sample to be used for later analysis.

3.3 Recording Procedure

Recordings took place in the fNIRS recording room in the Centre for Human Brain Health on the University of Birmingham campus in February 2026. The room was dimly lit to minimize any light leaking in to the headset. Participants were read a set of instructions about the recording and then led through a short example recording following a much shortened version of the protocol, ensuring that they understood the task. Only a single participant was permitted in the room at a time, and recordings were supervised throughout.

The protocol instructions were displayed on a standard Windows laptop running a custom Python script, which used PsychoPy for display design and timing and PyLSL for accurate time-marker transmission. This script would display the instructions, cues and would send the time-markers to the Gowerlabs software to store alongside the generated file. The instructions, Python code used, and recording details can be found in `/psychopy/` in the project repository.

3.4 Collected Dataset

Recordings were collected from 5 participants using the Gowerlabs LUMO headset across three two-hour recording sessions. Each participant completed one recording following the protocol described above.

All recordings were initially saved in LUFR format and were then converted to SNIRF format using LUMO's LUMOMAT MATLAB library. In some cases, the first event marker was not recorded correctly and was therefore added manually during data preparation. A full record of these modifications, including the augmentation code, is provided in `/data_augmentation/` within the project repository.

Although the full dataset was retained for inspection and preprocessing, its use for quantitative evaluation was limited by signal quality issues, which are discussed in the following subsection.

3.5 Limitations of the Data

Following collection, an initial visual inspection of the recordings suggested that the data was of acceptable quality. However, after processing the recordings through the full preprocessing pipeline and evaluating them using standard fNIRS analysis (block-averaging-based and GLM-based), the dataset did not show clear task-evoked activations at the expected times for any participant. Since this was inconsistent with the intended motor-task design, several stages of the processing code were reviewed and revised, and multiple analysis attempts were repeated to check whether the issue arose from the implementation rather than the recordings themselves.

To further investigate this, the recordings were also reviewed in discussion with the project supervisor and processed using an existing MATLAB-based fNIRS pipeline. This produced qualitatively similar waveform behaviour, suggesting that the lack of expected activation was not solely a consequence of the custom Python pipeline used in this project. One notable feature of the recordings was an unusually strong negative relationship between HbO and HbR, beyond what would typically be expected from a standard task-evoked response. While the exact cause could not be confirmed, this raised the possibility that some aspect of the recording process may have failed at either the hardware or software level.

As a result, the collected dataset was not considered sufficiently reliable for drawing strong conclusions about task-evoked activation recovery. It was still useful for testing and developing the processing pipeline, and parts of it will still be used in the evaluation, but its limitations should be recognised when interpreting the confidence in the evaluation results presented later in this report.

4 Methodology

4.1 Overview of the Processing Pipeline

A processing pipeline was developed to take a raw SNIRF file, recorded according to the protocol in `/psychopy/`, and transform it into corresponding HbO and HbR concentration time series. The pipeline has a wide range of tunable parameters across all stages of pro-

cessing, and returns a series of evaluation outputs for a defined set of correction pipelines on each of the three trials (NHM/SHM/LHM) in the recording.

Included is a flow chart of the pipeline (Figure 10)

Any signal frequency filters used in the processing pipeline use the same zero-phase third-order Butterworth implementation, using `scipy.signal.butter` and `filtfilt` to avoid phase distortion. Despite this, edge transients can still occur, and so 70 samples were trimmed from both ends of the signal after each filtering step. At the recording rate of 6.25 Hz, this corresponds to 11.2 s at each edge, and was chosen conservatively to reduce the effect of filter transients on later processing. The use of a third-order Butterworth filter is also consistent with common practice in the fNIRS literature and with the default filtering convention used in Homer2 [19].

4.2 Signal Preparation

4.2.1 Channel Selection

As mentioned previously, the output time series of the LUMO system is of shape $3456 \times [6.25 * t]$, meaning 3456 channels and a sampling rate of 6.25Hz over t seconds. While this is true, many of the channels are dropped prior to any processing.

Specifically, two subsets of channels are dropped before further processing: one because their nodes were not considered for analysis, and the other because their source-detector separations were either too short or too long to capture meaningful signal. In adult fNIRS channel distances can range from ≈ 8 mm to ≈ 50 mm, usually grouped into short-separation (8 – 15 mm) and long separation (20 – 50 mm).[20, 16, 1]. To exclude signals in the lower SNR range, the bounds 10 mm and 40 mm were chosen.

Performing this channel pruning on LUMOs 3456 channels, keeping only 6 of 12 nodes, and using $min = 10$ mm, $max = 40$ mm, reduces the number of channels to just 344. Figure 2 shows 3D plots of all 3456 channels and the reduced set of 344 channels after channel selection.

4.2.2 Conversion to Optical Density

Individually, each channel records the detected (raw) light intensity received from the source to the detector at the specified wavelength. Typically before any other processing, this is converted directly into Optical Density (OD) using the formula $OD(t) = -\log\left(\frac{I(t)}{I_0}\right)$, where I_0 is the mean signal amplitude over the initial recording baseline period. The optical density of a single channel can look like Figure 3. While some algorithms in fNIRS literature do perform other preprocessing on the raw light intensity, such as MA correction or frequency-filtering, none of the algorithms mentioned in this project consider it.

4.2.3 Signal Quality Evaluation

Additionally, before any correction is applied, signal quality is assessed in order to identify poor channels and to provide an indication of the overall recording quality. In the initial implementation, the intention was to discard all channels falling below a chosen signal-quality threshold. However, because the collected recordings were unusually noisy, this would have resulted in the removal of too many channels, potentially undermining the effectiveness of the multichannel correction methods. For this reason, the approach was changed so that signal quality was tracked and retained as a reference measure, rather than being used as a strict pruning criterion.

The signal quality metric used for this was the Scalp Coupling Index (SCI), computed on the baseline period, implemented following the PHOEBE paper [21]. SCI estimates channel quality by isolating the cardiac-frequency component of the raw intensity signals and measuring the correlation between the two wavelengths, with stronger agreement indicating better optical coupling and higher signal quality. Other signal quality metrics exist, including Peak Spectral Power (PSP), signal-to-noise ratio (SNR), signal quality index (SQI) and coefficient of variation (CV), but these were not used in this project. [21, 22]

4.3 Correction Pipelines

After signal preparation, the OD channels were passed through each of the correction pipelines considered in this project: **No Correction**, used as a baseline; **TDDR**, used as a single-channel comparison method; the proposed cluster-based GVTG-gated tPCA method, referred to as **GGtPCA**; and a variant of this method incorporating the GVTG channel filter, referred to as **fGGtPCA**. Each pipeline takes the prepared OD signal as input and returns a corrected OD signal for later conversion and evaluation.

4.3.1 No Correction

In this pipeline, no motion artefact correction is applied. The input OD signal is passed directly to the output, providing a reference for later analysis of the impact of the other correction pipelines.

4.3.2 TDDR

TDDR was used as the main single-channel comparison method. It was performed for each channel using the `tddr_pipeline`. The input OD signal was split into low- and high-frequency components using a low-pass cutoff of 0.5 Hz and both components were trimmed. TDDR was then applied only to the trimmed low-frequency component, and the corrected result was recombined with the corresponding high-frequency component to produce the corrected OD signal. In the full pipeline, this corrected signal was then returned with the trimmed edge regions reinserted so that the pipeline output retained the original signal length. In practice, this had no effect on later analysis, since the next stage of the pipeline applied a further frequency filter and trimmed the signal again. The purpose of trimming the edge transients prior to TDDR is to prevent them from influencing the estimated signal mean, which would in turn affect the output of the TDDR algorithm.

4.3.3 Cluster-Based GVTG-Gated tPCA Method (GGtPCA) Overview

The proposed multichannel correction pipeline is referred to as **GGtPCA**. In this pipeline, the input OD signal is first split into local channel clusters, after which GVTG is used within each cluster to identify motion-contaminated periods. These periods are then converted into correction bins, and tPCA is applied only to the affected parts of each cluster signal.

A variant of this method, **fGGtPCA**, additionally enables the optional GVTG channel filter before motion detection and correction. In this variant, GVTG bin calculation and tPCA correction are performed only on channels that pass the filter, after which all channels, including those identified as noisy, are concatenated back together to form the final output.

4.4 Detailed Structure of GGtPCA

4.4.1 Clustering

The purpose of clustering in this method is to allow motion detection and correction to be performed on local groups of channels rather than across the full recording at once. In a high-density system, nearby channels may share motion artefacts because they are affected by the same local optode movement, while more distant channels may be less related. Defining local clusters therefore provides a way to retain the multichannel nature of the correction step, while reducing the risk of combining unrelated channels into a single global correction.

Additionally, channels with different source-detector separations are not structurally equivalent, since their depth sensitivity differs and they can contain different proportions of superficial and cortical signal. Grouping channels with very different separations into the same multichannel correction step therefore risks combining signals with different underlying structure, which could cause issues during tPCA. Because of this, when defining clusters, the range of source-detector separations must be considered.

The goal of clustering, therefore, is to divide the full set of channels into distinct subsets such that each channel belongs to exactly one subset, with channels within each subset being spatially and structurally related. Two clustering methods are proposed in this project:

Node-Pair Clustering. The node-pair clustering algorithm forms clusters according to the pair of headset nodes to which a channel belongs. Under this scheme, a channel between nodes i and j is placed in the same cluster as a channel between j and i , so each cluster corresponds to a unique pair of hexagons on the LUMO headset. Given the 3-source, 4-detector configuration of a LUMO node, with 2 wavelengths measured for each source-detector pairing, this gives a maximum of $3 * 4 * 2 * 2 = 48$ channels in clusters where $i \neq j$ and $3 * 4 * 2 = 24$ channels where $i = j$.

The motivation for this approach is that motion artefacts in a channel are likely to originate from movement at one of its two contributing nodes, meaning that channels sharing the same node pair are likely to be affected by similar local motion. This produces spatially meaningful clusters that are directly tied to the physical headset structure.

A limitation of this clustering method is that the effective number of channels per cluster can become fairly small in practice. Although the maximum cluster sizes of 24 and 48 channels are reasonable, this number is reduced after excluding noisy channels and removing channels with overly short or long source-detector separations, leaving fewer channels for the later GVTG and tPCA stages.

K-Means Clustering. The K-means midpoint clustering algorithm instead groups channels according to the spatial location of their measurement midpoints. For each channel, the midpoint between its source and detector positions is computed in 3D space, and these midpoint coordinates are used as the feature vectors for K-means clustering. Given a chosen number of clusters k , the algorithm partitions the channels into k disjoint groups such that channels within each group have relatively nearby midpoints. This provides more flexibility in cluster size than the node-pair approach, since clusters are defined by proximity rather than strict headset topology. To avoid mixing channels with substantially different source-detector separations, the method first splits the channels into short-distance and long-distance groups using a distance threshold of 18 mm, and then applies K-means separately within each group. This ensures that channels grouped into the same correction step are not only spatially close, but also more similar in their measurement geometry.

An 18 mm threshold was used as a pragmatic split between shorter and longer source-detector separations, broadly consistent with the fNIRS literature, where short-separation channels have been used at 13-15 mm, 20 mm, and in some cases 25 mm [23].

Given a target average cluster size N_k , the number of clusters can be chosen as $k = \lfloor N/N_k \rfloor$. This helps avoid the issue of very small clusters that node-pair clustering suffers from. A disadvantage of K-means compared to node-pair clustering is that its clusters are not directly tied to the physical headset structure. It therefore trades some physical interpretability for greater flexibility in cluster size and spatial grouping.

4.4.2 GVTD-Based Motion Detection

For each cluster, GVTD-based motion detection was performed using the `gvtd_pipeline`. In the fGGtPCA variant, an initial GVTD calculation was first performed on the unfiltered cluster signal in order to apply the GVTD channel filter and identify noisy channels. While the original GVTD implementation uses a 60 s window to consider noisy channels, this implementation uses a window of 20 s to better line up with the block design used, as one can expect there to be multiple motion-free 20 s periods but few 60 s ones.

After optionally performing a channel filter, the main GVTD calculation was then performed on a temporally filtered and trimmed version of the retained cluster OD signal, using its own high-pass and low-pass cutoffs chosen specifically for motion detection rather than later signal processing. GVTD was then thresholded using the standard threshold rule with the tunable constant c . If no valid channels or no motion periods were found, the cluster was passed through unchanged.

Although the GVTD-tPCA process is iterated multiple times, in fGGtPCA the noisy channel mask is computed only once, before the first iteration, and is kept the same through each iteration.

4.4.3 Motion Bin Construction

The thresholded GVTD output was converted into a binary motion mask, from which contiguous motion bins were constructed. Each detected period was padded on both sides using a configurable temporal padding value so that the full motion-contaminated region, rather than only the threshold crossing itself, could be corrected. Since GVTD was computed on the filtered and trimmed signal, the resulting bins were then mapped back to the indexing of the original cluster signal before correction was applied.

4.4.4 Targeted PCA Correction

If a cluster contained at least one valid channel and at least one motion bin, tPCA was applied to the selected channels within that cluster. Only the samples falling inside the detected motion bins were used in this step, so that correction was targeted specifically at the motion-contaminated periods rather than the full signal. The binned segments were concatenated in time, PCA was applied to this concatenated signal, and the leading components were removed according to the configured variance target and maximum number of principal components.

After tPCA correction, the corrected bins were written back into their original temporal positions in the cluster signal. This reconstruction step follows the same general idea used in standard spline interpolation and later tPCA implementations, where corrected motion segments are shifted back into the full signal in a way that preserves continuity at the bin boundaries [9, 10, 2]. In the implementation used here, this was done by aligning each corrected bin to the mean signal immediately before and after the bin using a short alignment window, and then applying a small taper at both bin edges to smooth the transition back into the uncorrected signal. Channels not selected for correction were left unchanged.

4.4.5 Reconstruction of the Corrected Signal

For each cluster, only the channels selected for correction were updated, while all other channels were left unchanged. After all clusters had been processed independently, their corrected signals and metadata were concatenated back together to form the final corrected OD output returned by the pipeline.

4.5 Post-Correction Processing

After MA correction there are still a few steps before the data can be analysed using the various evaluation methods. These include a temporal filter, conversion from OD to HbO and HbR concentrations, and splitting the recording into the three individual trials (NHM, SHM, LHM).

4.5.1 Band-Pass Filtering

After correction, the corrected OD data is passed through a band-pass filter using a frequency band of 0.01 Hz – 0.2 Hz. These cutoffs were selected with the block design in mind. As mentioned, each block is 30 s long (10 s task + 20 s rest). This corresponds to a

fundamental frequency of approximately $\frac{1}{30} \approx 0.033$ Hz. A suitable pair of cutoffs should retain the energy of this frequency while suppressing lower-frequency baseline drift or higher-frequency physiological noise.

The low cutoff of 0.01 Hz is a common choice in task-based fNIRS preprocessing for removing very slow baseline drift that can originate from different hardware or physiological sources [19, 2]. This corresponds to a period of 100 s, which is much slower than our 30 s cycle, and so it should not have a significant effect on the task-related response. The high cutoff of 0.2 Hz was chosen to retain task-related responses and their low-order harmonics (0.066 Hz, 0.133 Hz) while reducing any faster physiological fluctuations, such as respiration (0.2 – 0.6 Hz) and cardiac pulsation (0.6 – 2.5 Hz) [24].

It should be noted that Mayer waves, a set of low-frequency oscillations in arterial blood pressure that manifest in fNIRS signals at approximately 0.1 Hz, fall within this passband and therefore cannot be removed without also attenuating the task-related signal [25]. This represents an inherent limitation of the chosen filter design, as any cutoff aggressive enough to suppress Mayer waves would risk distorting the haemodynamic response of interest (which is also expected occur at 0.1 Hz).

4.5.2 Modified Beer-Lambert Law Conversion

Finally, the OD signal is converted into HbO and HbR concentrations using the *Modified Beer-Lambert Law* (MBLL) [19]. This conversion requires a couple parameters, most importantly a table of wavelength-dependent HbO/HbR extinction coefficients. In this project, Scott Prahl’s haemoglobin extinction coefficient table was used, in line with common fNIRS software implementations such as Homer2 [26, 27]. Since the raw-to-OD conversion uses the natural logarithm via `np.log()` and channel distances are measured in mm, while Prahl’s extinction coefficients are conventionally tabulated in decadic form and cm-based units, the coefficients were scaled by $\ln(10)/10 \approx 0.2303$ before conversion. This is consistent with the unit conversion used in common fNIRS software implementations such as MNE-NIRS [28]. The remainder of the MBLL step followed a standard two-wavelength implementation, producing paired HbO and HbR concentration time series for each source-detector channel.

4.5.3 Trial Slicing

In order to compare the three trials in each recording fairly, all preprocessing is done across the entire recording rather than on individual slices. This kept a fairer comparison when looking at the impact of the motion artefact correction algorithms in the context of the entire processing pipeline. Now that the signals have been processed into their final form (HbO/HbR) they should be split in order to be evaluated independently.

The trial slicing was simple to perform. Each trial was defined to start at the beginning of its preceding baseline period and to end at the end of its following baseline period. This meant that every extracted trial contained both its own task-rest blocks and the baseline segments immediately surrounding them. Since the baseline periods between consecutive trials were shared, the extracted trial windows overlapped slightly at their boundaries. This was intentional, as it allowed each trial to retain a complete local baseline context for later evaluation.

4.6 Unused Processing Methods

Superficial Signal Regression (SSR) and global signal regression are commonly used in fNIRS preprocessing to remove systemic physiological noise. SSR uses dedicated short-separation channels (< 10 mm) as regressors for long-separation channels [22], while global regression subtracts the mean across all channels, assuming that globally shared components are non-neural [12].

In practice, both methods can also suppress motion-related artefacts when these are shared across channels. For this reason, neither approach was part of the intended pipeline. The aim of this project is to evaluate motion artefact (MA) correction methods, so applying regressions that may themselves remove or attenuate motion artefacts would interfere with that comparison.

However, once the data quality issues described in Section 3.5 became clear, both methods were trialled in simplified forms in an attempt to recover a usable signal. Global regression was straightforward to apply, whereas SSR was more involved, particularly due to choices around channel pairing and the signal domain used for regression. Given time constraints, more careful implementations were not pursued.

In practice, neither method led to any meaningful improvement. This aligns with the overall data quality. Both approaches assume that an underlying neural signal can be recovered once shared noise is removed. With most channels showing poor scalp coupling, that assumption does not hold, so there is effectively little signal for either method to recover.

4.7 Implementation Details

The full processing and evaluation pipeline was implemented in Python 3.11, using NumPy and SciPy for numerical computation, pandas for channel metadata management, and matplotlib for visualisation. Rather than building on an existing fNIRS toolbox such as Homer2 or MNE-NIRS, all processing and correction methods were implemented from scratch. This choice was made to ensure full transparency over every stage of the pipeline and to allow the flexibility needed when developing a custom correction method.

Repository Structure. The pipeline code is organised into small, single-purpose modules under `src/`. The three main orchestration scripts live in `src/pipelines/`: the processing pipeline, the correction pipelines, and the evaluation pipeline, which together form the end-to-end processing chain. Evaluation methods are separated into `src/evaluation/`, and each processing step and correction algorithm has its own module under `src/`. Further detail on the repository layout is given in Section 7.

SNIRF File Handling. As mentioned, recordings were saved in the LUFR format and converted to SNIRF prior to processing. The `snirf_processing` module handles loading these files, extracting the raw intensity array, channel metadata, sampling rate, and stimulus markers into easier to use formats (dictionaries, NumPy arrays).

Correction Pipeline Architecture. The correction stage is built around a class hierarchy designed to make each method modular and interchangeable. A base `CorrectionPipeline` class defines a common interface: it accepts a standardised `CorrectionPipelineIn` dataclass containing the OD signal, channel metadata, sampling rate, and parameter dictionary, and returns a `CorrectionPipelineOut` dataclass with the corrected signal and associated metadata. Each concrete correction method (`NoCorrection`, `TDDR`, `GGtPCA`, `fGGtPCA`) subclasses this base and implements its own pipeline method. Adding a new correction method to the comparison therefore requires only writing a new subclass and passing it to the pipeline.

Signal Representation Conventions. Raw and OD signals are stored as arrays of shape $(2 * n_{\text{channels_per_wavelength}}, n_{\text{samples}})$, while wavelength-paired data uses $(2, n_{\text{channels_per_wavelength}}, n_{\text{samples}})$, where the first axis indexes the two wavelengths (or HbO and HbR after MBLL conversion). Both representations were retained because different stages of the pipeline operate on different views of the data: OD calculation and filtering act on individual channels, SCI computation and MBLL conversion require wavelength-paired channels, and different correction methods can require different combinations. Channel metadata is maintained in a paired `pandas DataFrame` whose rows correspond to the channel axis. Wavelength pairing is handled through a `couple_signals_fn` function built from the channel metadata, which returns the correctly matched channel pairs and a corresponding boolean mask.

Overall Data Flow. The end-to-end pipeline takes a single SNIRF file and a shared parameter dictionary as input to the `processing_pipeline`. From this input, the processing pipeline loads the recording, applies the selected preprocessing and correction methods, and returns a Python dictionary containing the processed data products needed for later analysis. This includes both the overall recording after processing and the individual trial recordings derived from it, with intermediate sub-processing outputs still retained, such as OD signals, filtered signals, channel-filtered metadata, and final HbO/HbR concentration signals. This processed dictionary is then used by the evaluator, which applies the chosen evaluation methods using the stored outputs. The evaluator returns a second dictionary containing the resulting evaluation summaries, metrics, and any supporting method-specific outputs for every combination of trials and correction pipelines. The same parameter dictionary is passed throughout this process, so that processing, correction, and evaluation all operate under a single explicit set of settings.

Verification of Reimplemented Methods. Since all processing and correction methods were implemented from scratch, each major component - including band-pass filtering, TDDR, GVTD, tPCA, clustering, and SNIRF loading - was verified individually during development. This was done by developing and implementing each algorithm in isolation (refer to the various notebooks in `src/notebooks/`), applying it to recorded data, and visually inspecting the outputs at each stage to confirm that the method produced the expected effect on the signal. For example, TDDR was checked to confirm whether the TD distribution was truly repaired, GVTD was confirmed to peak during visually obvious motion periods, and tPCA was confirmed to reduce variance in the identified motion bins without visibly distorting clean segments. Intermediate outputs and signal behaviour were also reviewed in discussion with the project supervisor, who has prior experience with fNIRS processing. While this visual verification approach does not provide formal numerical equivalence with existing toolbox implementations, it was applied consistently across all components and provided sufficient confidence that each stage behaves as intended.

4.8 Default Parameters

A single default parameter set was defined for the processing and correction pipeline. Since the dataset was too limited for meaningful hyperparameter optimisation, the large majority of parameters were fixed throughout and should be understood as implementation defaults for this project. Table 3 lists the values used for preprocessing, TDDR, and GGtPCA.

Among these parameters, the most important design choices were the GVTD threshold constant c , the clustering method, the target cluster size, and the tPCA variance target. The constant c controls how aggressively motion-contaminated periods are detected, so it has the clearest effect on the balance between under-correction and over-correction. The clustering method and target channels per cluster determine the locality and scale of the multichannel correction step. The variance target controls how much of the motion-bin variance

is removed during tPCA, with 0.97 chosen to match common practice in the tPCA literature [2]. The remaining parameters were kept fixed for consistency across recordings and methods, rather than as part of the main experimental comparison.

5 Evaluation

5.1 Intended Evaluation Methods

The primary goal of the evaluation was to determine whether MA correction could recover a plausible task-evoked haemodynamic response from recordings contaminated by instructed head motion. Two standard fNIRS evaluation approaches were implemented for this purpose: block-averaged responses and a General Linear Model (GLM). Both methods rely on the assumption that the underlying data contains a recoverable task-related signal, and in both cases the expected result would be a characteristic increase in HbO concentration and a corresponding decrease in HbR concentration in channels over the motor cortex during the finger-tapping task periods.

5.1.1 Block-Averaged HbO and HbR

Block averaging is a common method for evaluating task-evoked fNIRS responses [19]. The HbO and HbR time series for each channel are segmented into their individual task-rest blocks, each aligned to task onset, and then averaged across blocks. Under the assumption that the task-evoked response is time-locked to onset and that noise averages to zero over trials, the resulting waveform isolates the haemodynamic response. For a clean motor task recording, one would expect to observe a positive HbO peak and a negative HbR trough emerging a few seconds after task onset, peaking around 5-6 s and persisting through the 10 s task window. In a well-functioning evaluation, block-averaged responses could be compared across the NHM, SHM, and LHM trials and across the four correction pipelines (No Correction, TDDR, GGtPCA, fGGtPCA) comparing the average amplitudes, the variability in signal, and the HbO/HbR relationship. A correction method that successfully removes motion artefacts would be expected to produce a cleaner, more consistent block-average that more closely resembles the NHM response in the SHM and LHM trials.

5.1.2 General Linear Model

The GLM is a model-based approach to identifying task-related activation in fNIRS data [19]. A design matrix is constructed by convolving the task onset regressor with a canonical haemodynamic response function (HRF), producing a predicted response time course. This is then regressed against the measured HbO and HbR signals at each channel. The resulting beta coefficients indicate the strength of the task-related response, and t-statistics can be used to assess statistical significance. Channels showing significant positive beta values for HbO and negative values for HbR over the motor cortex region would constitute evidence of successful activation detection.

5.2 Data Quality and Its Impact

Before presenting the evaluation results, it is necessary to talk about the quality of the collected recordings, as this has a direct bearing on which evaluation methods can be applied and how confidently conclusions can be drawn.

5.2.1 Signal Quality Assessment

Signal quality was assessed using the Scalp Coupling Index (SCI) computed over the initial baseline period of each recording. SCI measures the correlation between the two wavelengths at the cardiac frequency. In the PHOEBE implementation, a score above 0.8 indicates a clean channel [21], but given an initial inspection showed poor channel quality, this threshold was lowered to 0.5 in this project, hoping to increase the number of viable channels. Table 1 reports the number of channels in the filtered set of 344 that meet this threshold for each participant.

Four of the five participants have effectively no usable channels under this threshold, with three below 1.5%. Unfortunately, Subject 1 is the only participant with a meaningful proportion of clean channels at 20.9%, and is therefore the sole participant considered in the evaluation that follows.

5.2.2 Failure of Standard Evaluation Approaches

GLM and block-average analysis were applied to Subject 1's NHM trial under no correction as a ground truth verification step.

Block-averaging was first restricted to the 10 highest-SCI channels with source-detector separations of 25 mm or greater whose midpoints fell within 40 mm of the estimated motor cortex ROI (Figure 5a). The separation constraint ensures sufficient depth sensitivity to reach cortical tissue, while the proximity constraint targets the region where a contralateral finger-tapping response would be expected. The resulting waveforms show no clear task-evoked response. The standard error is high and the mean trace contains repeated oscillatory

peaks at a frequency just below 0.2 Hz, likely due to some physiological signal surviving the bandpass filter. Additionally, the SCI of channels in this region ranges only from 0.2-0.4, confirming extremely poor optical coupling in the area where a motor response would be expected. Further inspection of the spatial distribution of SCI scores (Figure 4a) confirmed that no channels near the motor cortex ROI were of high quality, with the highest-SCI channels being more concentrated towards the frontal region of the headset.

Approaching from the opposite direction, manually inspecting individual channels and selecting those with the most plausible block-averaged waveforms revealed that such channels were spatially scattered across the array rather than clustering over motor cortex, which is not consistent with a localised cortical response. This implies that these signals are more likely caused by random fluctuations than by genuine task-related response.

Relaxing the spatial constraint and selecting the 10 highest-SCI channels globally (Figure 5b) produced a more structured waveform, but one in which HbO and HbR were positively correlated, both rising and falling together rather than exhibiting the anti-correlated pattern expected from HbO/HbR pairs. This was traced to the fact that these channels have near-identical optical density signals across the two wavelengths in the haemodynamic band, with correlation coefficients averaging $r = 0.86$. When both wavelengths carry effectively the same signal, the MBLL inversion cannot separate HbO from HbR and instead produces two outputs that both track the shared component.

To account for this, the channel selection was repeated with an additional constraint excluding channels whose inter-wavelength OD correlation exceeded 0.83 (Figure 5c). For a derivation of this threshold, see Appendix A. This recovered the expected anti-correlated relationship between HbO and HbR, but the response itself remained weak and noisy.

The GLM told the same story, showing no coherent activation pattern over the expected region.

A response that only appears on spatially dispersed channels and disappears when restricted to the motor cortex is not a motor cortex response. The time-locked structure that does appear is more likely global physiological noise intertwined with the task timing than a genuine neural signal. Since this persists in the cleanest recording condition, it reflects a fundamental signal quality limitation rather than something correctable. The likely causes are discussed in Section 3.5. Consequently, task-based evaluation cannot be applied, and an alternative approach is adopted in Section 5.3.

5.3 Alternative Evaluation Approach

Because the dataset was not adequate for validating recovery of true task-evoked haemodynamics, the final evaluation focuses on whether the methods reduce motion-like corruption while preserving plausible signal structure, rather than whether they recover statistically convincing motor activation.

This is a significant limitation. Task-evoked recovery is the standard basis for evaluating correction methods, and without it the strongest form of validation is unavailable. However, the core contribution of this project is the design and implementation of a cluster-based correction method, and whether a method reduces motion-related corruption can still be assessed to some extent without a known underlying response [13, 6], even if the conclusions that can be drawn are necessarily weaker.

Additionally, because the evaluation is no longer bound to the motor cortex, the node restriction applied during pipeline design (Section 4.2.1) is no longer necessary. For the analysis that follows, all 12 nodes were included, expanding the channel set from 344 to 744 after distance-based pruning, increasing the statistical power for the metrics used below. Given this new larger set of channels, the channels signal quality was re-evaluated in Table 2, showing, once again, that Subject 1's recording is the only one suitable for evaluation. The spatial distribution of SCI scores for Subject 1 under the new expanded channel set is shown in Figure 4b.

The following evaluation uses three metrics, each targeting a different aspect of correction quality. The first assesses whether correction suppresses motion-related contamination during instructed motion periods. The second assesses whether correction preserves the expected anti-correlated relationship between HbO and HbR, a basic physiological property of clean fNIRS signals [8]. The third checks whether correction introduces unnecessary distortion to periods where no motion was instructed. Together, these cover artefact removal, signal preservation, and over-correction. None of them can determine whether the corrected signal is closer to some unknown ground truth, only whether it is more plausible, and this limitation should be kept in mind when interpreting the results that follow.

5.3.1 Artefact Suppression: Event-Locked GVTD Attenuation

The purpose of this metric is to assess whether each correction pipeline reduces the motion-related disturbance that was deliberately introduced during the SHM and LHM trials. Because the onset times of all instructed motion events were recorded alongside the fNIRS data, the impact of each event on the signal can be measured directly by examining the GVTD trace in a short window around the onset.

GVTD was chosen as the basis for this metric because it has been shown to correlate strongly with external measures of motion and to have high sensitivity and specificity to instructed motion events [1].

For each correction pipeline p , a single GVTD trace $g_p(t)$ is computed from the corrected and band-passed OD signal. Both wavelengths are included and the standard GVTD formula is applied.

For each instructed motion event e with onset time τ_e , a local pre-motion baseline is first estimated as the median GVTD in the interval before the event:

$$B_{p,e} = \text{median}(g_p(t)), \quad t \in [\tau_e - 5, \tau_e - 1]$$

The half-second gap before onset avoids including any preparatory movement that may precede the instructed motion. The motion burden for that event is then defined as the integrated excess GVTD above this baseline over the first 5 seconds after motion onset:

$$M_{p,e} = \Delta t \sum_{t \in [\tau_e, \tau_e + 5]} \max(g_p(t) - B_{p,e}, 0)$$

where $\Delta t = 1/f_s$ is the sampling interval. The $\max(\cdot, 0)$ term ensures that only positive excursions above the local baseline contribute to the score, so that the metric captures excess motion-related activity rather than the overall signal level.

For each trial condition (SHM or LHM), the per-event scores are aggregated using the median across the 8 motion events in that trial:

$$S_{p,\text{trial}} = \text{median}_e M_{p,e}$$

The median is used rather than the mean to reduce the influence of any single event with an unusually large or small response. The final reported quantity is the percentage attenuation relative to the uncorrected pipeline:

$$A_{p,\text{trial}} = 100 \times \left(1 - \frac{S_{p,\text{trial}}}{S_{\text{NoCorrection},\text{trial}}} \right)$$

An attenuation of 0% indicates no improvement over the uncorrected signal, while 100% would indicate complete removal of the event-locked motion burden. Negative values would indicate that correction increased the motion burden relative to no correction.

This metric is computed only for the SHM and LHM trials, as the NHM trial contains no instructed motion events and therefore no meaningful motion onsets to lock the analysis to. The impact of correction on the NHM trial is instead assessed by the other two evaluation metrics.

It should be noted that this metric is not entirely neutral with respect to the proposed GGtPCA method, since GGtPCA uses GVTD internally as part of its motion detection step. The implications of this are examined in Section 5.5.4, and the second and third evaluation metrics, which are independent of GVTD, provide a counterbalance.

5.3.2 Signal Preservation: HbO-HbR Anti-Correlation

The purpose of this metric is to assess whether each correction pipeline preserves the expected anti-correlated relationship between HbO and HbR. While the first metric targets artefact suppression, this metric targets signal preservation: a correction method that distorts the underlying signal structure, for instance by projecting out components that carry genuine physiological variance, would be expected to degrade this relationship even if it successfully suppresses motion-related contamination.

For each correction pipeline p and trial condition, the Pearson correlation coefficient between the HbO and HbR concentration time series is computed independently for each channel j that passes the quality filter:

$$\rho_{p,j} = \frac{\sum_t (\Delta \text{HbO}_{j,t} - \overline{\Delta \text{HbO}_j}) (\Delta \text{HbR}_{j,t} - \overline{\Delta \text{HbR}_j})}{\sqrt{\sum_t (\Delta \text{HbO}_{j,t} - \overline{\Delta \text{HbO}_j})^2 \sum_t (\Delta \text{HbR}_{j,t} - \overline{\Delta \text{HbR}_j})^2}}$$

where the sums run over all time points t in the segment of interest. This is computed over three temporal segments: the full trial, the concatenated task periods only, and the concatenated rest periods only. Since instructed motion was only introduced during the task periods, the rest periods provide a set of clean segments that can be used as a within-trial reference for all correction methods.

Before computing the metric, channels are filtered to exclude those with low SCI or high inter-wavelength OD correlation, as discussed in Section 5.2.2. As shown there, the highest-SCI channels tend to be dominated by channels whose two wavelengths carry near-identical signals, producing positively correlated HbO and HbR outputs as an artefact of the MBLL inversion rather than as a consequence of correction quality. If these channels are not excluded, they skew the resulting distributions and obscure any effect of the correction methods. Channels are therefore retained only if they satisfy both $\text{SCI} > \theta_{\text{SCI}}$ and $r_{\text{OD}} < r^*$, where $r^* \approx 0.83$ is the threshold derived in Appendix A.

The resulting per-channel correlation values $\{\rho_{p,j}\}$ form a distribution for each pipeline, trial, and segment combination. These distributions can be compared across pipelines using summary statistics such as the median, interquartile range, and the fraction of channels exhibiting the expected negative correlation, as well as through visual inspection of the full distributions. Ideally, a successful correction method would shift the distribution towards stronger negative correlations in the SHM and LHM task periods [8], where motion contamination is expected to have degraded the relationship, while leaving the distribution largely unchanged in the NHM trial and in rest periods where no motion was introduced.

It should be noted that this metric cannot distinguish between a genuine physiological anti-correlation and one arising from other sources of shared variance. However, since the same ambiguity applies equally across all four correction pipelines, relative comparisons remain informative.

5.3.3 Over-Correction: Rest-Period Signal Variability

The purpose of this metric is to assess whether each correction pipeline over-corrects periods of the signal where no correction should be needed. While the first metric evaluates artefact suppression and the second evaluates signal structure preservation, neither captures a method that uniformly flattens the signal. *GVTD Attenuation* would register such flattening as successful suppression, while *HbO/HbR Anti-Correlation* would not detect it at all, since Pearson correlation is scale-invariant. This metric addresses that gap directly.

For each correction pipeline p , trial condition, and channel j that passes the SCI quality filter, the standard deviation of the HbO and HbR concentrations are independently computed over the concatenated rest-period samples (1). The primary output is the per-channel percent change in standard deviation relative to the uncorrected pipeline (2):

$$(1) \sigma_{p,j} = \text{std}(\Delta \text{HbO}_{j,t}), \quad t \in \text{rest periods} \quad (2) \Delta \sigma_{p,j} = 100 \times \frac{\sigma_{p,j} - \sigma_{\text{NoCorr},j}}{\sigma_{\text{NoCorr},j}}$$

A negative value indicates that the correction method reduced the variability of that channel during rest periods; a positive value indicates it increased it.

Rest periods are the ideal window for this check because they should contain minimal motion contamination, especially in the NHM trial where no motion was instructed at any point. Any reduction in variability during NHM rest periods cannot be attributed to successful artefact removal and must therefore represent over-correction. In the SHM and LHM trials, a modest reduction may be acceptable if residual effects of instructed motion bled into adjacent rest periods.

Unlike *HbO/HbR Anti-Correlation*, the inter-wavelength OD correlation filter is not applied here, since this metric does not depend on the MBLI inversion's ability to separate HbO from HbR. It measures only the amplitude of fluctuations, which is meaningful regardless of whether the two wavelengths are well-separated.

On its own, this metric has limited interpretive value. Preserving rest-period variability does not prove that a method corrected motion artefacts effectively, since a method that does nothing at all would score perfectly, but combined with *GVTD Attenuation*, a good result here confirms that the attenuation was targeted rather than indiscriminate.

5.4 Results

As discussed in Section 5.2, the following results are based only on Subject 1, since the remaining recordings were too noisy to support a meaningful evaluation. Unless otherwise stated, all analyses use the default parameter set reported in Table 3, together with an SCI threshold of 0.5 when determining which channels to include. Given the extremely limited dataset, the results should be interpreted as indicative rather than conclusive, and are used primarily to examine relative trends and trade-offs between methods.

5.4.1 SHM as the Primary Motion Condition

Before presenting the sensitivity analysis and final pipeline comparisons, it is necessary to clarify the role of the two motion conditions used in the evaluation. Although both SHM and LHM contained instructed motion, the two conditions did not contribute equally informative evidence in Subject 1's recording.

As shown in Figure 6, the uncorrected SHM trial produced a much clearer relationship between instructed motion intervals and peaks in the GVTD trace than the uncorrected LHM trial. This suggests that, in practice, the LHM trial did not generate a comparably strong or reliable motion signature in this recording, despite being intended as the larger-motion condition, or at least not one captured by the GVTD metric.

This difference was also reflected in the parameter selection and later pipeline comparisons, where attenuation values in LHM varied erratically across configurations and often failed to track the clearer trends seen in SHM. For this reason, SHM was treated as the primary motion condition when assessing motion suppression, while LHM was retained only as a secondary reference.

5.4.2 Sensitivity Analysis of (f)GGtPCA Parameter Choices

To choose a final parameter set for GGtPCA and fGGtPCA, and to understand how the main design choices influenced the results, a sensitivity analysis was carried out over the GGtPCA and fGGtPCA parameters. The sweep covered 12 configurations for each pipeline. For node-pair clustering, three values of the GVTD threshold constant were tested ($c \in \{3.5, 4.5, 5.5\}$). For midpoint K-means clustering, the same three c values were tested alongside three target cluster sizes (40, 80, and 120 channels per cluster). All other parameters were kept fixed at their default values.

Selection Criteria. The main quantity used to assess the sweep was the SHM event-locked GVTD attenuation (**SHM att.**), for which larger values indicate stronger suppression of the instructed motion response. Signal preservation was assessed primarily using the NHM full-trial HbO-HbR correlation metrics, reported as the change in median correlation relative to No Correction (**NHM $\Delta \rho_{\text{full}}$**) and the change in the fraction of negatively correlated channels (**NHM Δf_{full}^-**), together with the average absolute change in NHM rest-period

signal variability ($\text{NHM} |\Delta\sigma_{\text{rest}}|$). Three secondary supporting metrics were also examined in SHM: the change in task-period median correlation ($\text{SHM} \Delta\rho_{\text{task}}$), the change in the fraction of negatively correlated channels ($\text{SHM} \Delta f_{\text{task}}^-$), and the average absolute change in SHM rest-period signal variability ($\text{SHM} |\Delta\sigma_{\text{rest}}|$). The first two indicate whether correction shifts the motion-contaminated task periods towards a more plausible HbO–HbR relationship, with a more negative $\text{SHM} \Delta\rho_{\text{task}}$ and a more positive $\text{SHM} \Delta f_{\text{task}}^-$ being preferable. The third checks whether this improvement is achieved without excessive distortion of the adjacent rest periods, for which smaller values of $\text{SHM} |\Delta\sigma_{\text{rest}}|$ are preferable. The results for all tested configurations are summarised in Table 4.

Observed Trends. The clearest overall pattern in the sweep was that fGGtPCA consistently achieved much higher **SHM att.** than GGtPCA, with values of approximately 53%–73% compared with 3%–25% for the unfiltered variant. Since the purpose of the sweep was to select one configuration within each method family, the more relevant comparisons are those made within GGtPCA and within fGGtPCA separately.

Within GGtPCA, *MidpointKMeans* generally outperformed *NodePair* on **SHM att.**, and the 40-channel K-means configurations were usually weaker than the 80- and 120-channel settings. Lower values of c also tended to give stronger attenuation, although the difference between the strongest K-means settings was relatively small. The *NodePair* configurations were notably more variable in the preservation metrics: at $c = 3.5$, node-pair clustering produced the highest NHM rest-period variability distortion of any configuration tested ($\text{NHM} |\Delta\sigma_{\text{rest}}| = 6.9\%$), while achieving only 15% SHM attenuation, making it the clearest over-aggressive GGtPCA configuration in the sweep.

Within fGGtPCA, attenuation remained strong across a much wider range of configurations, but the NHM preservation metrics became more important for distinguishing between them. More aggressive settings (lower c , or node-pair clustering) often gave slightly higher **SHM att.** but at the cost of larger NHM distortion. For instance, node-pair clustering at $c = 3.5$ achieved the highest SHM attenuation (71%) but also one of the largest NHM rest-period distortions ($\text{NHM} \Delta f_{\text{full}}^- = 0.167$, $\text{NHM} |\Delta\sigma_{\text{rest}}| = 2.5\%$).

Selected Configurations. For GGtPCA, *MidpointKMeans* with 120 channels per cluster and $c = 4.5$ was selected. It achieved one of the strongest SHM attenuation values within the unfiltered family (24.3%), with a favourable NHM anti-correlation shift ($\text{NHM} \Delta\rho_{\text{full}} = -0.070$, $\text{NHM} \Delta f_{\text{full}}^- = 0.083$) and essentially zero measurable NHM rest-period standard deviation change. For fGGtPCA, *MidpointKMeans* with 80 channels per cluster and $c = 5.5$ was selected. Although this did not maximise SHM attenuation (65.6%), it achieved the smallest NHM distortion of any fGGtPCA configuration ($\text{NHM} \Delta\rho_{\text{full}} = -0.007$, $\text{NHM} |\Delta\sigma_{\text{rest}}| = 0.047\%$) while still maintaining strong SHM task-period anti-correlation improvement ($\text{SHM} \Delta\rho_{\text{task}} = -0.219$). Both selected configurations are highlighted in Table 4.

5.4.3 Comparison Against Baseline Pipelines

Using the parameter settings selected in Section 5.4.2, the final comparison was carried out between **No Correction**, **TDDR**, **GGtPCA**, and **fGGtPCA**. The comparison focuses on three qualities of each pipeline: artefact suppression, signal preservation, and over-correction. Since all reported quantities are defined relative to the uncorrected signal, **No Correction** serves only as the baseline reference.

Artefact Suppression. On the primary suppression metric, **SHM att.**, the four pipelines were clearly separated. TDDR achieved by far the largest attenuation (90.9%), followed by fGGtPCA (65.6%) and GGtPCA (24.3%). The SHM supporting metrics showed a similar pattern, but with fGGtPCA substantially closer to TDDR. TDDR and fGGtPCA both shifted the SHM task-period HbO–HbR relationship in the expected direction, with $\text{SHM} \Delta\rho_{\text{task}}$ of -0.228 and -0.219 respectively, and $\text{SHM} \Delta f_{\text{task}}^-$ of 0.167 for both. GGtPCA also improved these quantities, but much more modestly ($\text{SHM} \Delta\rho_{\text{task}} = -0.088$, $\text{SHM} \Delta f_{\text{task}}^- = 0.042$). These results indicate that TDDR was the most aggressive suppressor of SHM motion contamination, fGGtPCA removed a substantial proportion of it, and GGtPCA was comparatively conservative.

Although **LHM att.** is not included in the main summary table (as mentioned in Section 5.4.1), it is worth noting that TDDR produced a much larger attenuation in LHM than fGGtPCA (63.6% vs 5.3%). However, as illustrated in Figure 7b, this appears to reflect the broader effect of TDDR on GVTd peaks throughout the trial rather than a clearer motion-locked suppression.

Figure 7a illustrates the difference visually. TDDR reduces the GVTd trace across a much larger proportion of the signal, including smaller fluctuations outside the motion-locked spikes, while fGGtPCA leaves more of the surrounding structure intact and acts primarily on the larger peaks. The mechanisms underlying this difference are discussed in Section 5.5.3.

Signal Preservation. The signal-preservation metrics again separated TDDR from the two multichannel methods more clearly than they separated GGtPCA from fGGtPCA. In the clean NHM condition, where values closer to zero are preferable, fGGtPCA remained closest to the NoCorrection reference ($\text{NHM} \Delta\rho_{\text{full}} = -0.007$, $\text{NHM} \Delta f_{\text{full}}^- = 0.000$), while GGtPCA also produced only a modest shift (-0.070 , 0.083). TDDR behaved very differently, with $\text{NHM} \Delta\rho_{\text{full}} = +0.252$ and $\text{NHM} \Delta f_{\text{full}}^- = -0.208$, indicating a much larger

alteration of the clean-condition HbO-HbR relationship. The SHM rest-period correlation metrics showed the same overall pattern, with GGtPCA and fGGtPCA remaining close to the baseline while TDDR produced the largest deviation.

Figure 8 makes this difference clearer by showing the per-channel correlation-change distributions for TDDR and fGGtPCA. For fGGtPCA, the distribution remains tightly concentrated around zero, with only limited spread into more negative and more positive values. This is consistent with a selective correction method that leaves most channels close to the No Correction reference while still altering a smaller subset. By contrast, TDDR shows a much flatter and more widely spread distribution, with substantial mass extending into the positive side and no strong peak at zero. This indicates that TDDR alters the HbO-HbR relationship across a much larger proportion of channels, rather than leaving most channels close to the clean-condition baseline.

Over-correction. The rest-period variability results again separated TDDR clearly from the two multichannel methods. In the NHM condition, where no correction should ideally occur, GGtPCA showed practically zero change in rest-period variability, while fGGtPCA remained very close to the NoCorrection reference (**NHM HbO** = -0.013% , **NHM HbR** = $+0.081\%$). By contrast, TDDR produced large negative changes in both chromophores (**NHM HbO** = -15.689% , **NHM HbR** = -5.272%), indicating substantial attenuation of the clean-condition signal. The same pattern was even more pronounced in SHM rest periods, where TDDR again showed very large negative shifts (**SHM HbO** = -17.112% , **SHM HbR** = -38.614%), while fGGtPCA produced much smaller changes (**SHM HbO** = -1.644% , **SHM HbR** = -0.679%) and GGtPCA again remained at zero. The fact that this effect is more pronounced during SHM suggests that it is at least partly due to leakage from the motion artefacts into the adjacent rest periods.

Figure 9 supports this interpretation visually. In both NHM and SHM, the TDDR rest-period HbO variability distribution is shifted well to the left of the No Correction reference, and the corresponding per-channel percentage changes are broadly spread across strongly negative values. By contrast, fGGtPCA remains much more concentrated near the uncorrected distribution, with per-channel changes clustered close to zero.

Overall Comparison. The three evaluation metrics paint a consistent picture. TDDR achieved the strongest artefact suppression, but at a substantial cost to signal preservation and rest-period variability. GGtPCA produced minimal distortion, but was too conservative to remove a meaningful proportion of the motion contamination. fGGtPCA offered the most favourable trade-off, achieving substantial suppression together with task-period correlation improvements close to those of TDDR, while preserving the clean-condition signal structure and introducing only minimal over-correction.

5.5 Discussion

5.5.1 Overall Outcome of the Project

The project delivered what it set out to build: a modular Python pipeline implementing the full processing chain from raw SNIRF files to evaluated HbO/HbR outputs, with each processing and correction stage implemented transparently from scratch, including the proposed GGtPCA and fGGtPCA methods and the baseline TDDR method. Together with the experimental recordings and evaluation framework, this constitutes a complete and reproducible system for investigating multichannel motion correction in dense fNIRS data.

What the project could not deliver was a strong empirical validation. The data quality limitations described in Sections 3.5 and 5.2 meant that standard task-recovery evaluation was unavailable, and the proxy metrics used in its place can assess whether a corrected signal is more plausible but not whether it is more accurate. This limits how confidently differences between methods can be interpreted, and should be kept in mind throughout the discussion that follows.

Despite these limitations, the indicative results from the single usable recording are striking. fGGtPCA achieved 65.6% attenuation of the instructed motion response while leaving the clean-condition signal almost entirely unchanged. Its task-period correlation improvements were close to those of TDDR, but without TDDR's substantial cost to rest-period variability and clean-condition signal structure. The fact that a gated, cluster-local method can approach the suppression performance of a well-regarded single-channel method while preserving signal properties that TDDR measurably degrades is a promising result.

The remainder of this discussion examines what these results reveal about the design choices underlying GGtPCA, the nature of the trade-offs between the three correction approaches, and the limits on what can be inferred from the available evidence.

5.5.2 What the Results Suggest About the GGtPCA Design

The most consistent pattern in the sensitivity analysis was the large performance gap between fGGtPCA and GGtPCA. This gap can be attributed to two related mechanisms. First, without the channel filter, GVTD is computed over the full set of cluster channels, many of which have poor optical coupling and contribute substantial noise to the temporal derivative. This inflates the GVTD baseline, making genuine motion events less distinct relative to the surrounding signal and causing fewer motion periods to exceed the detection threshold. Second, when tPCA is subsequently applied to clusters containing these noisy channels, a larger proportion of the total variance is attributable to noise rather than coherent motion-related structure, meaning the variance removed by the leading principal components may not correspond well to the shared motion artefact that tPCA is designed to target. The channel filter addresses both

problems: by excluding noisy channels before GVTD computation, motion events become more prominent in the GVTD trace, and by excluding them from the tPCA step, the variance captured by the leading components is more likely to reflect genuine motion structure.

This interpretation is supported by an interaction between the channel filter and the threshold constant c . The selected configurations used $c = 4.5$ for GGtPCA and $c = 5.5$ for fGGtPCA, meaning the filtered variant achieved stronger attenuation with a less aggressive threshold. Across both methods, lower values of c consistently increased SHM attenuation but also increased the risk of over-correction, visible in the NHM signal preservation metrics. The fact that fGGtPCA could operate at a higher c while still achieving strong suppression suggests that the channel filter and threshold constant are not independent design choices but compensate for each other.

The clustering results also revealed meaningful patterns. Within both GGtPCA and fGGtPCA, midpoint K-means clustering generally outperformed node-pair clustering, and node-pair configurations showed more erratic behaviour in the preservation metrics, with the $c = 3.5$ node-pair case producing the worst NHM rest-period distortion of any configuration tested (6.9%). There are several likely reasons for this. Node-pair clustering is limited to maximum cluster sizes of 24 or 48 channels depending on whether the source-detector pair shares a node, which after quality-based pruning can leave clusters with very few usable channels for GVTD and tPCA to operate on. By contrast, K-means clustering allows direct control over cluster size through the target channels-per-cluster parameter. Additionally, node-pair clusters can mix channels with substantially different source-detector separations, since two nodes can have both very short and very long inter-node channels. This conflicts with the assumption discussed in Section 2.3.1 that channels in the same tPCA correction step should share similar measurement characteristics. K-means clustering partially mitigates this by first partitioning channels into short-separation and long-separation groups using an 18 mm threshold before clustering within each group, though it should be noted that this threshold was a fixed design choice that was not itself tested and could influence the results. Node-pair clustering often outperformed K-means with 40 channels per cluster despite operating at similar effective cluster sizes. This suggests that the physical node structure does retain some useful information about which channels share motion-related behaviour, even if the overall approach is less flexible than K-means.

Among the K-means configurations, 40-channel clusters were consistently weaker than 80 and 120 in both GGtPCA, reinforcing the pattern that very small clusters may not contain enough channels for GVTD to reliably distinguish motion from noise or for tPCA to identify a stable motion-related component. In fGGtPCA this effect was less pronounced but still apparent, likely because the channel filter had already removed the noisiest channels, giving even smaller clusters a better signal-to-noise ratio for both detection and correction.

Some parameters were kept fixed throughout the evaluation due to time constraints and should be acknowledged as unexplored. The tPCA variance target of 0.97 was inherited from Yücel et al. [2] and was not varied during the evaluation. Whether this value is appropriate for data of this quality is difficult to assess without task-based evaluation, since the existing metrics can measure over-correction in rest periods but cannot determine whether tPCA removed genuine signal within the corrected motion bins themselves. Similarly, the bin padding (0.8 s), number of iterations (3), alignment window, and taper length were fixed at values chosen during development. Informal testing during development suggested that realistic variations in these parameters did not dramatically alter the output, but they were not subjected to the same systematic sweep as c , clustering method, and cluster size. However, the pattern of fGGtPCA outperforming GGtPCA was stable across all configurations that were tested, which provides confidence in the relative ranking of methods if not in the specific values reported.

5.5.3 Trade-offs Between Correction Approaches

The results from Section 5.4.3 show clear differences between the three correction pipelines, which can be traced to fundamental properties of each method's design.

TDDR's strong suppression and its cost to signal preservation both stem from the same mechanism. It operates independently on each channel and downweights all large temporal-derivative outliers without distinguishing their source, meaning it catches artefacts that a threshold-based detector would miss, but cannot leave clean segments of the signal intact. Figure 7a illustrates this, with TDDR dampening peaks throughout the trial rather than only at motion-locked times. The NHM rest-period variability changes confirm that this effect extends beyond motion-contaminated periods.

GGtPCA sat at the opposite extreme. As discussed in Section 5.5.2, this is not a fundamental limitation of the method's design but rather evidence that unfiltered GGtPCA requires a level of channel quality that this dataset did not provide.

fGGtPCA occupied the middle ground, substantially weaker than TDDR in raw attenuation but much stronger in preservation behaviour. The gap between fGGtPCA's attenuation and TDDR's likely reflects a fundamental difference between gated and ungated correction. fGGtPCA can only correct motion events that produce a clear GVTD exceedance within the cluster, so smaller artefacts that fall below the detection threshold are missed entirely. TDDR operates on the full temporal derivative distribution and can downweight less prominent outliers that would not trigger a binary detection step. Figure 7a supports this, with both methods reducing the major motion-locked spikes but TDDR additionally suppressing smaller fluctuations that fGGtPCA leaves untouched.

This distinction also maps onto a broader difference between single-channel and multichannel correction. TDDR treats each channel independently and makes no use of spatial relationships, whereas fGGtPCA exploits the spatial structure of high-density systems through clustering, multichannel detection, and multichannel correction. The existing tPCA literature has noted that effectiveness depends

heavily on the motion detection step [2, 9], which is typically single-channel. The results presented here, while only indicative, suggest that a multichannel detection step can capture instructed motion events effectively.

Huang et al. [6] concluded that no single correction method is universally best. The results of this project are consistent with that conclusion, with TDDR and fGGtPCA representing genuinely different trade-offs rather than one being strictly better than the other. Which is preferable will depend on the application, the recording quality, and whether the downstream analysis is more sensitive to residual motion contamination or to signal distortion.

5.5.4 Limits of Inference and Threats to Validity

The results are drawn from a single participant, and the patterns observed could be idiosyncratic to this particular recording's noise characteristics and spatial distribution of channel quality rather than reflecting general properties of the methods. The implications of the absent task-recovery validation were discussed in Section 5.5.1, but it is worth examining how this affects each metric individually. The GVTD attenuation metric is not entirely neutral with respect to fGGtPCA, which uses GVTD internally for detection, while the preservation and over-correction metrics are inherently more sensitive to TDDR's global dampening than to a gated method's localised correction. The fact that all three metrics, despite these opposing biases, support a consistent relative ordering of methods provides some reassurance that the overall pattern is not purely an artefact of metric choice.

The evaluation also compared only three correction approaches. TDDR was a well-motivated baseline given its strong reported performance and shared use of the temporal derivative domain, but the fNIRS literature contains a wide range of other methods, both gated and ungated, single-channel and multichannel. Comparing fGGtPCA against additional methods such as wavelet-based correction, MARA, or CBSI would have provided a clearer picture of where it sits among existing approaches and strengthened the basis for any conclusions drawn.

It is also worth considering whether the data quality amplified fGGtPCA's advantage. The channel filter is most beneficial when many channels have poor coupling, so the gap between GGtPCA and fGGtPCA may narrow in higher-quality recordings. However, noisy channels are common across fNIRS recordings in practice and are routinely excluded before analysis, so the filter addresses a recurring problem rather than one specific to this dataset.

The recording protocol introduces further caveats. The 1-second instructed motion events were brief and occurred only once per 30-second block, consistent with the infrequent-motion assumptions underlying both GVTD and TDDR. In a setting with more frequent or sustained motion, TDDR's assumptions about the temporal derivative distribution would be more directly challenged, as the robust weighting step assumes that most time points are motion-free. For GVTD, more frequent motion would shift the distribution positively, potentially raising the threshold for a given value of c since the threshold is derived from the distribution mode. This would require a lower c to maintain the same detection sensitivity, though the effect depends on whether the mode itself is displaced, which would only occur if motion-contaminated periods became the majority. The trial ordering may also be relevant: since the NHM, SHM, and LHM trials were always recorded in that fixed order, the LHM trial may have been affected by cumulative changes in headset fit from the preceding trials, which could partly explain why it produced a weaker motion signature than expected despite being intended as the larger-motion condition.

5.5.5 Future Work

The most immediate priority is a proper empirical and statistical validation. This would require higher-quality recordings from multiple participants with confirmed task-evoked responses in the NHM condition so that task recovery can serve as the primary evaluation criterion. Future data collection should prioritise more thorough signal quality checks during cap fitting, particularly for participants with thicker hair, which is a likely contributor to the poor optical coupling observed in this study.

On the methodological side, several extensions follow naturally from the gaps identified in this evaluation. The GVTD threshold constant c was fixed per-recording, but an adaptive scheme that adjusts c based on the local noise characteristics of each cluster could reduce the need for manual tuning. The tPCA variance target and other fixed parameters identified in Section 5.5.2 would also benefit from systematic exploration. The cluster-and-detect framework is not tied to tPCA specifically, and testing whether alternative correction methods could be used within the same gated, cluster-local structure would help determine whether the benefit observed here comes from the detection strategy, the correction method, or their combination. It would also be valuable to reimplement the pipeline using established fNIRS toolboxes such as Homer2 or MNE-NIRS. While building from scratch was necessary for the transparency and flexibility required to develop fGGtPCA, integrating with existing libraries would make it substantially easier to compare against the wider range of correction methods already available in those toolboxes.

The K-means clustering approach is not inherently tied to the LUMO headset and could be applied to any high-density fNIRS system with sufficient channel density without modification. Node-pair clustering could be adapted to other node-based headset designs but would require a system-specific implementation for each. Testing the method on data from other systems would be an important step in assessing its generalisability.

6 Timeline

The first semester was spent building an understanding of fNIRS as a modality and exploring where within the field a project could contribute. Over the course of the semester, my focus narrowed towards motion artefact correction, with the spatial structure of the LUMO headset standing out as an underexplored advantage over the single-channel methods that dominate the existing literature. This period also involved deep reading of existing correction methods to inform the design of my own approach.

Development began at the start of the second semester. The specific decision to combine GVTd with tPCA in a cluster-based framework came during this period, as the overall pipeline took shape and the strengths of each method became clearer.

Data collection did not take place until February, and the data quality issues described in Section 3.5 were discovered soon after, leaving limited time to diagnose the issue, redesign the evaluation strategy, implement alternative metrics, and carry out the sensitivity analysis and final comparison.

7 Repository

The project repository is organised as follows. All Python source code lives under `src/`, with subfolders `pipelines/`, `evaluation/`, and `plots/` for their respective purposes. The correction pipelines themselves are in `pipelines/correction_pipeline/`, and the remaining processing stages are implemented as individual modules directly under `src/`.

Two notebook folders are included. `notebooks_results/` contains the final evaluation outputs: `sensitivity_analysis.ipynb` and `results_analysis.ipynb` reproduce the tables presented in this report, while `pipeline_view.ipynb` allows the full pipeline to be traced from start to finish with intermediate plots. `notebooks/` contains earlier exploratory work carried out during development, including experimentation with individual processing stages and parameters.

`psychopy/` contains the full recording protocol, including the PsychoPy script used to run the experiment, the instructions presented to participants, and a summary of the recording details. `data/` holds the SNIRF recordings, `data_augmentation/` contains the scripts used to add missing stimulus markers as described in Section 3.4, and `files/` contains the haemoglobin extinction coefficients used in the MBL conversion.

`notes/` contains informal notes collected throughout the project. Within it, `notes/plots/` holds all figures generated for the report, and `notes/information.ipynb` contains early but detailed documentation of the SNIRF file structure produced during the initial exploration phase.

Figures and Tables

Participant	Channels ≥ 0.5 SCI	Percentage
Subject 1	72 / 344	20.9%
Subject 2	38 / 344	11.0%
Subject 3	2 / 344	0.6%
Subject 4	0 / 344	0.0%
Subject 5	4 / 344	1.2%

Table 1: Number and percentage of channels with SCI ≥ 0.5 considering only the node and distance-pruned channel set.

Participant	Channels ≥ 0.5 SCI	Percentage
Subject 1	170 / 744	22.8%
Subject 2	46 / 744	6.2%
Subject 3	26 / 744	3.5%
Subject 4	6 / 744	0.8%
Subject 5	10 / 744	1.3%

Table 2: Number and percentage of channels with SCI ≥ 0.5 considering only the distance-pruned channel set across all nodes.

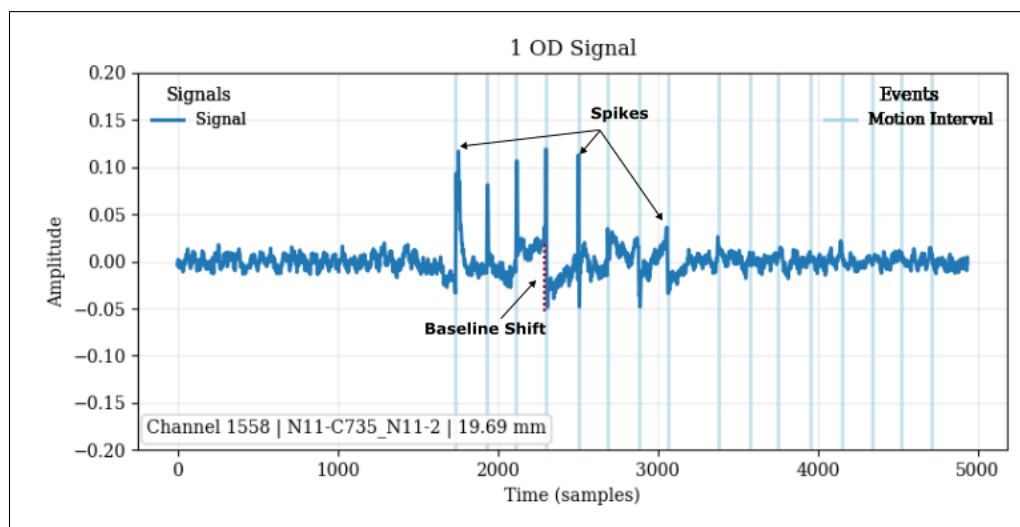


Figure 1: Examples of motion artefacts in the optical density (OD) signal, including spikes and baseline shifts. The OD signal was low-pass filtered at 1 Hz to enhance the visibility of these artefacts. Channel label is formatted "Channel index | Channel source-detector label | Channel source-detector separation".

Category	Parameter	Default	Source / Reasoning
Preprocessing	highpass	0.01 Hz	Pinti et al. [19]
Preprocessing	lowpass	0.2 Hz	See Section 4.5.1
Preprocessing	trim.samples	70	See Section 4.1
Preprocessing	min.dist	10 mm	See Section 4.2.1
Preprocessing	max.dist	40 mm	See Section 4.2.1
Preprocessing	keep.nodes	None	See Section 5.2.1
Preprocessing	use.baseline	True	Standard OD definition; this work
TDDR	tddr.lowpass	0.5 Hz	Fishburn et al. [3]
TDDR	tddr.trim.samples	70 samples	See Section 4.1
TDDR	tddr.max.iter	50	Fishburn et al. [3]
GGtPCA	c	4.5	Sherafati et al. [1]; this work
GGtPCA	bin.padding	0.8 s	This work
GGtPCA	gvtd.highpass	0.02 Hz	Sherafati et al. [1]
GGtPCA	gvtd.lowpass	0.5 Hz	Sherafati et al. [1]
GGtPCA	gvtd.trim.samples	70 samples	See Section 4.1
GGtPCA	gvtd.filter.window	20 s	See Section 4.4.2
GGtPCA	var.target	0.97	Yücel et al. [2]
GGtPCA	max.pcs	None	This work
GGtPCA	align.win	4 samples	This work
GGtPCA	taper	3 samples	This work
GGtPCA	ggtpca.iter	3	Yücel et al. [2]
GGtPCA	cluster.method	MidpointKMeans	This work
GGtPCA	channels.per.cluster	80	This work

Table 3: Default preprocessing and correction parameters used unless otherwise stated. Evaluation-specific parameters are omitted here and introduced later with the corresponding evaluation methods. It's worth noting that the `c`, `cluster.method` and `channels.per.cluster` parameters are explored in this project, and so their defaults are arbitrary choices.

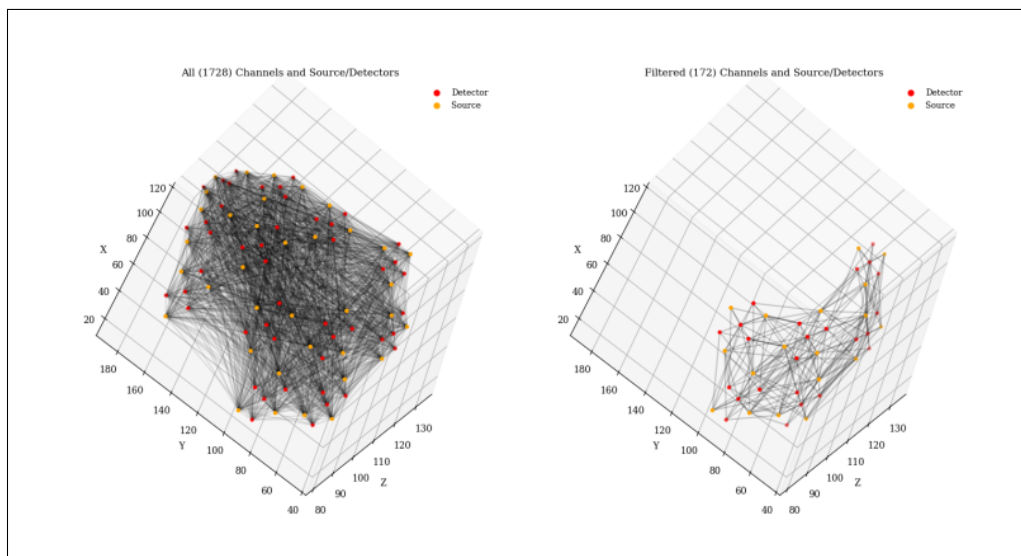


Figure 2: 3D view of source, detectors, and channels before and after the channel filter. While there are 3456 and 344 channels remaining, there are two for each source-detector pair, so only half of them are plotted

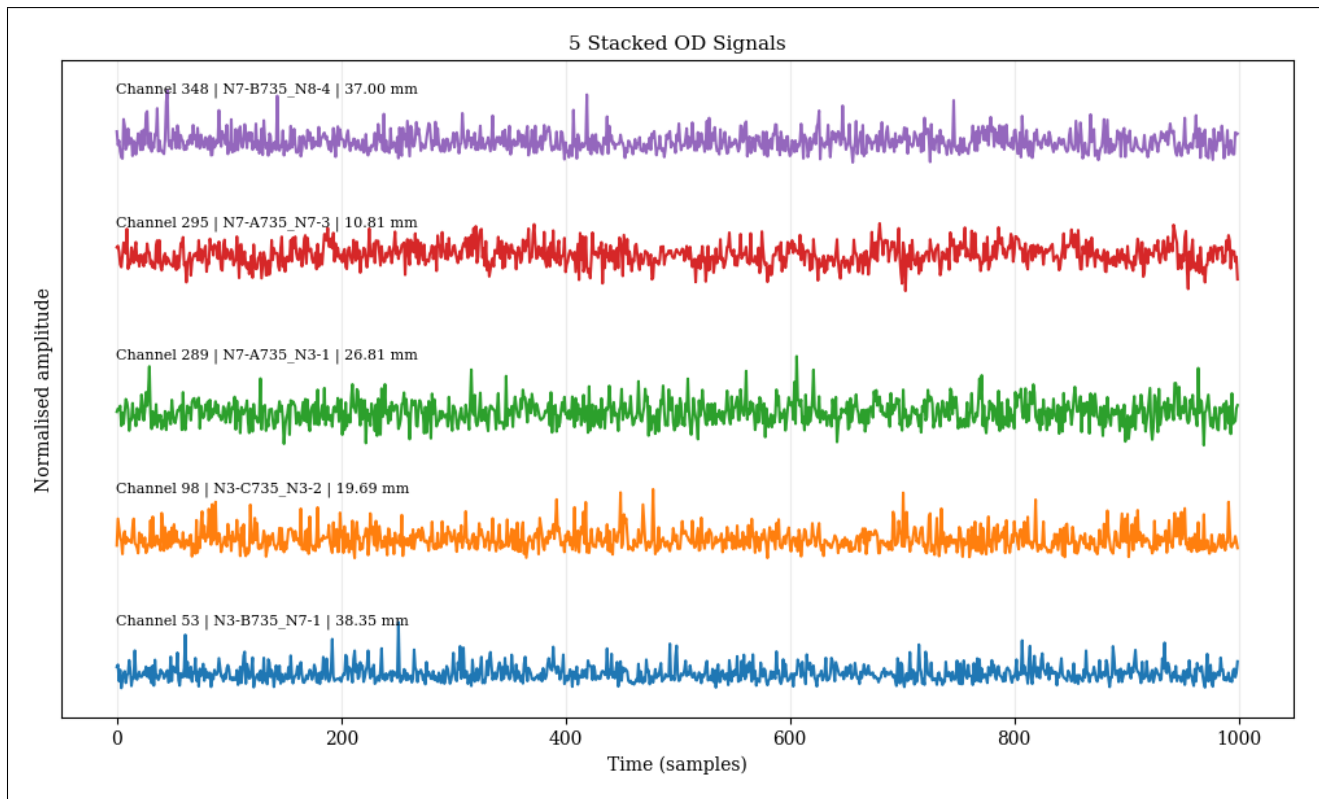
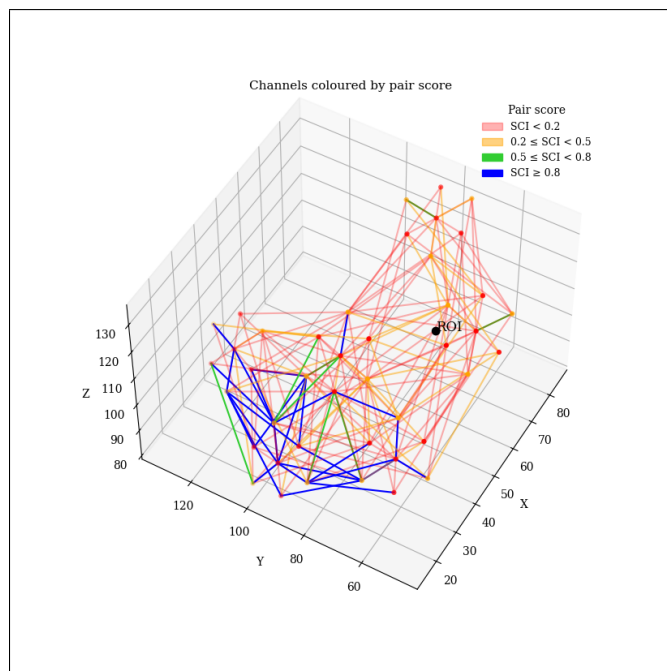
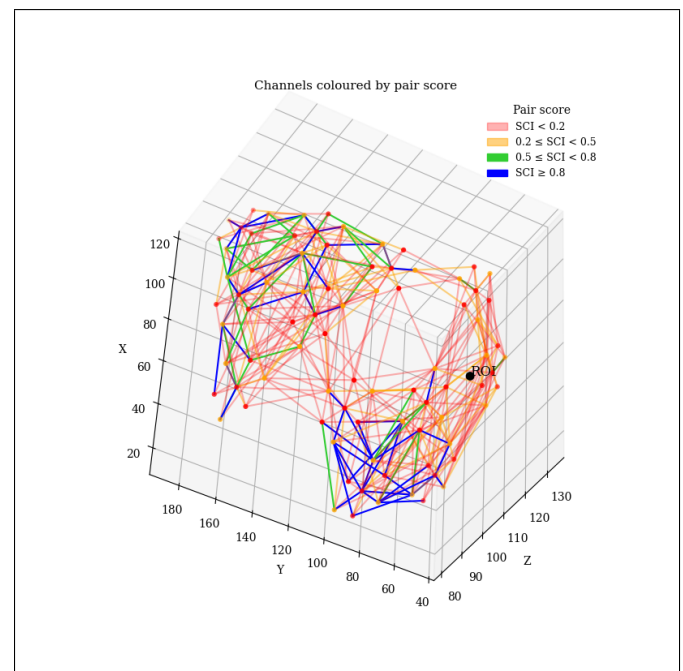


Figure 3: Examples of optical density (OD) signals. Channel label is formatted "Channel index | Channel source-detector label | Channel source-detector separation".



(a) All 344 channels with source-detector separations of $10 \leq d \leq 40$ mm across the 6 nodes selected for the original pipeline design.



(b) All 744 channels with source-detector separations of $10 \leq d \leq 40$ mm across the full 12-node array.

Figure 4: 3D view of channels coloured by SCI values.

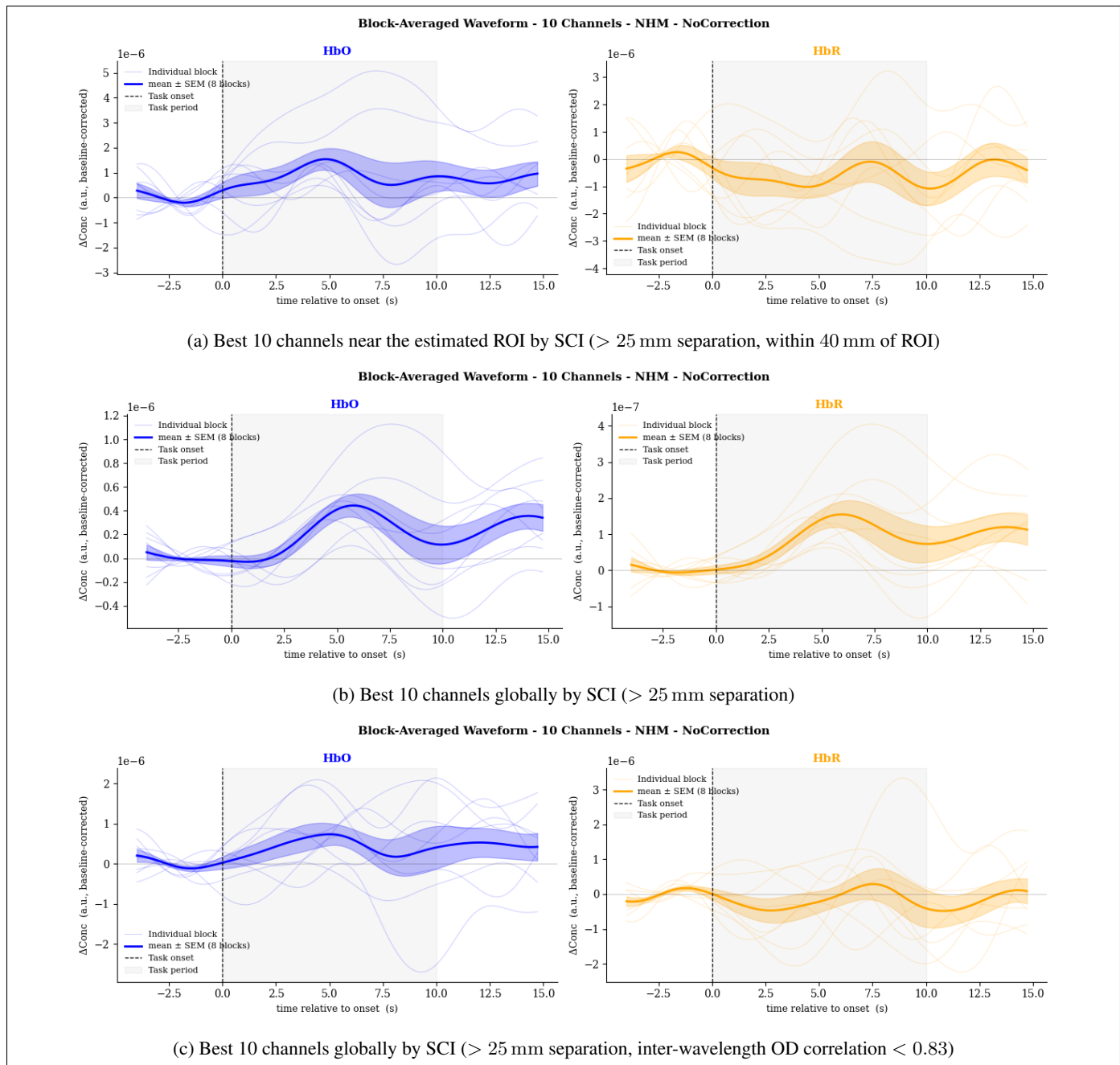


Figure 5: Block-average responses for Subject 1's NHM trial under no correction, under three channel-selection strategies. (a) Channels near the ROI show high SEM and no discernible response. (b) The highest-SCI channels globally produce a structured waveform, but HbO and HbR are positively correlated. (c) Excluding channels with high inter-wavelength OD correlation recovers the expected anti-correlated pattern, but the response remains weak and spatially dispersed.

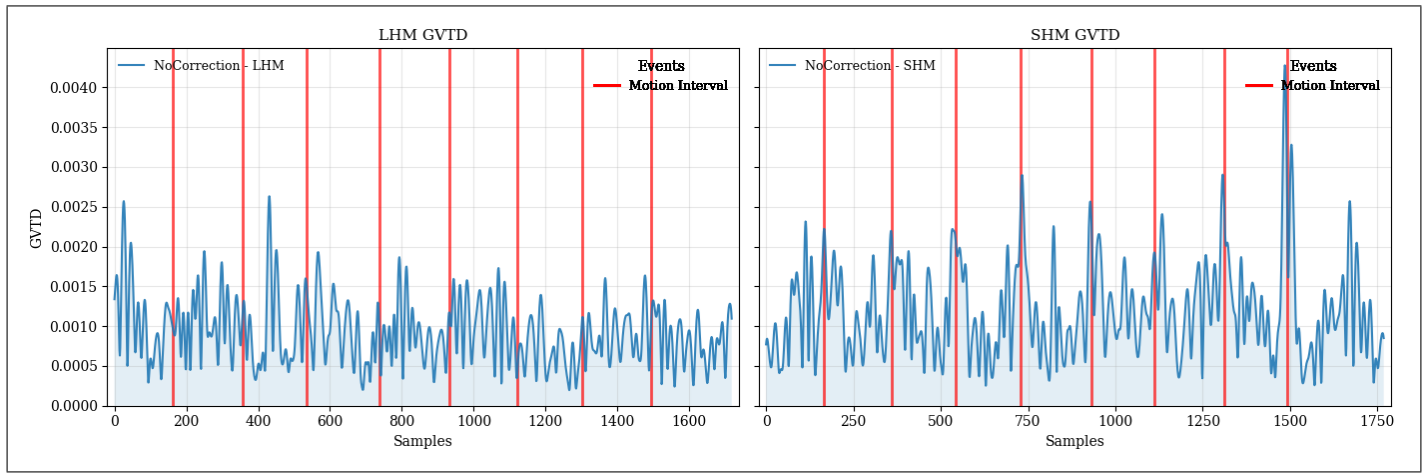
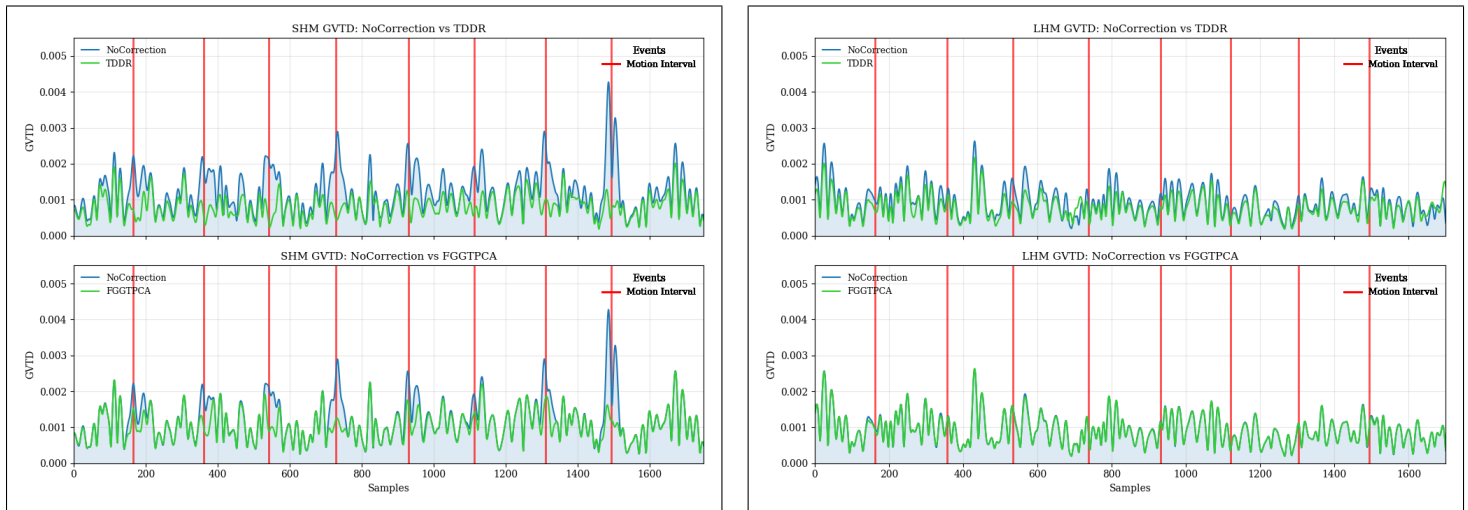


Figure 6: GVTD of the band-passed OD signal for the SHM and LHM trials under no correction, shown with a shared y-axis. In SHM, GVTD spikes align clearly with the instructed motion intervals, whereas in LHM this relationship is weaker and less consistent.



(a) GVTD in the SHM trial. Notably, in TDDR the GVTD reduction is widespread among the signal, whereas in fGGtPCA the GVTD reduction is localised to the larger spikes.

(b) GVTD in the LHM trial. Notably, TDDR has a dampening effect on the majority of peaks in the signal - whether they line up with motion intervals or not - whereas fGGtPCA has little effect at all, explaining the difference in **LHM Att.**

Figure 7: GVTD traces comparing TDDR and fGGtPCA with NoCorrection in the SHM and LHM trials, with motion induction intervals marked in red.

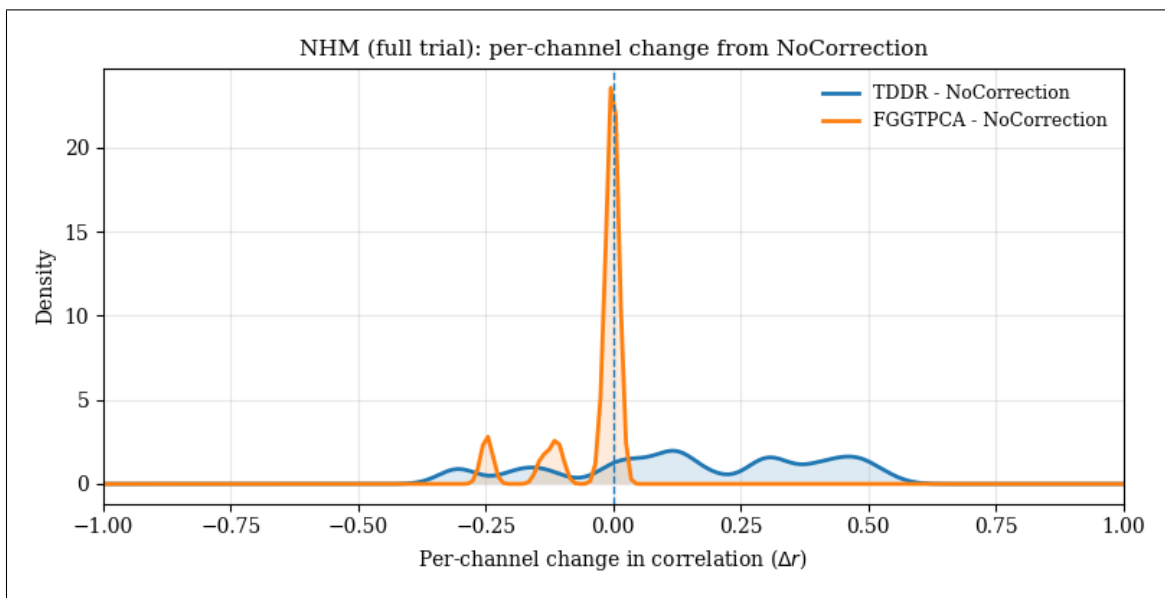


Figure 8: Kernel density estimate of the per-channel change in NHM full-trial HbO-HbR correlation relative to No Correction for TDDR and fGGtPCA. fGGtPCA remains strongly concentrated around zero, whereas TDDR is spread more broadly and is less centred on the uncorrected reference.

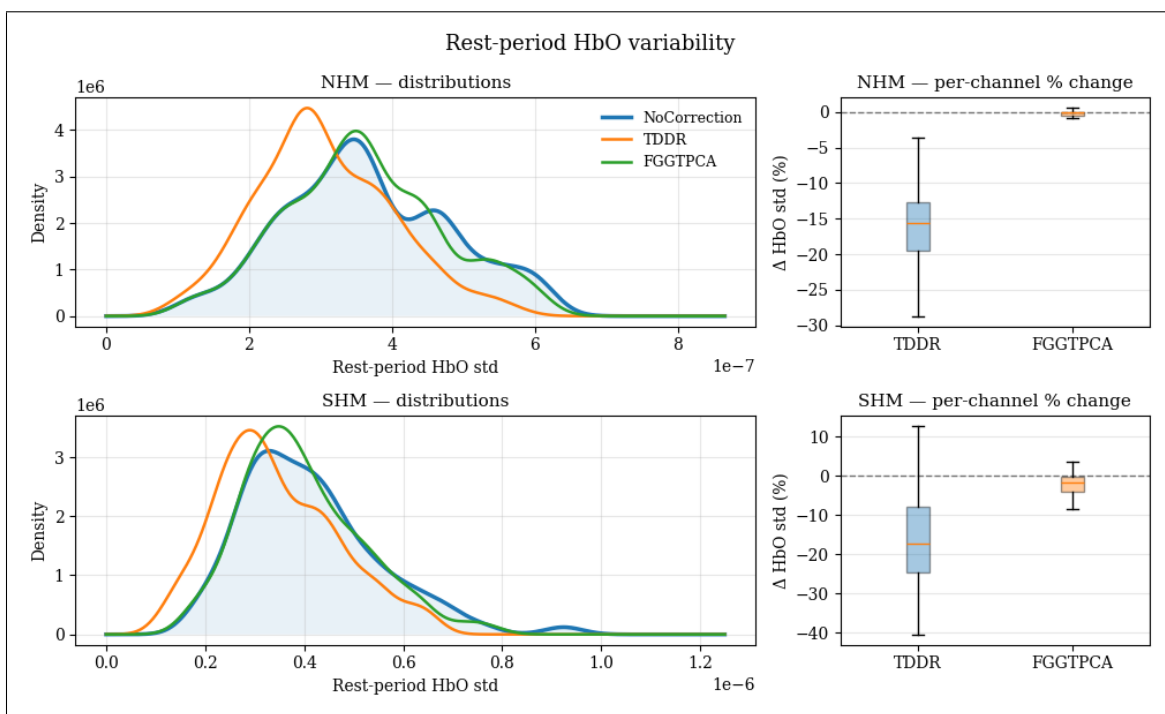


Figure 9: Rest-period HbO variability under No Correction, TDDR, and fGGtPCA for NHM and SHM. Left panels show the distributions of per-channel rest-period HbO standard deviation, and right panels show the corresponding per-channel percentage change relative to No Correction. TDDR shifts the variability distribution leftward and produces broadly negative per-channel changes in both conditions, whereas fGGtPCA remains much closer to the uncorrected reference.

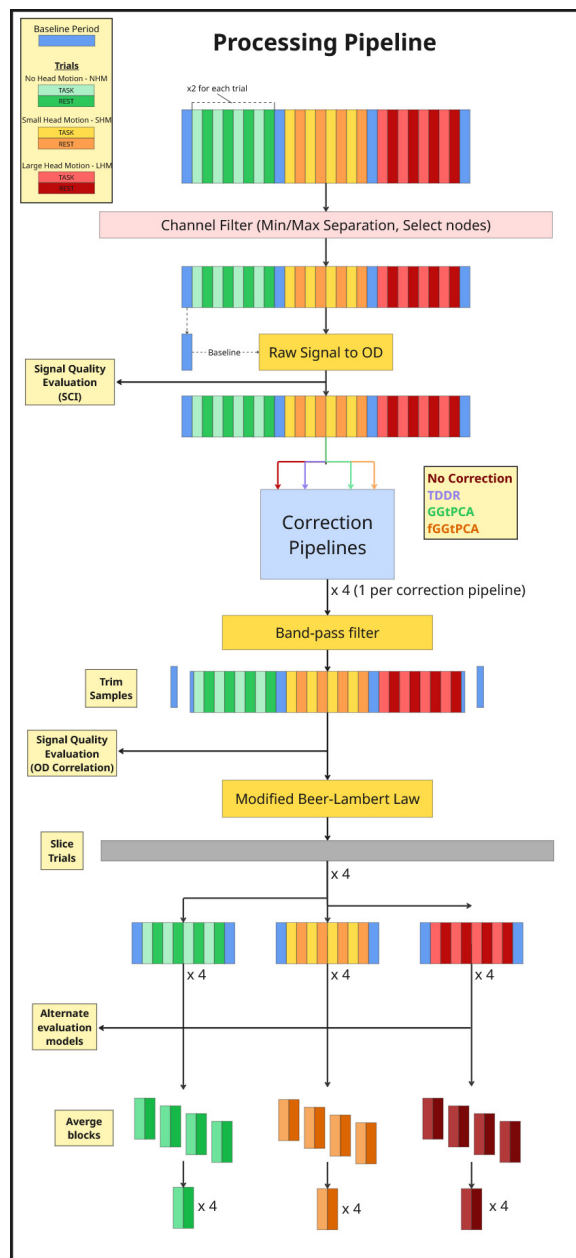


Figure 10: Flowchart of the processing pipeline.

Pipeline	Clustering	N_k	c	Primary metrics				Secondary metrics		
				SHM att. (%)	NHM $\Delta\rho_{full}$	NHM Δf_{full}^-	NHM $ \Delta\sigma_{rest} $ (%)	SHM $\Delta\rho_{task}$	SHM Δf_{task}^-	SHM $ \Delta\sigma_{rest} $ (%)
fGGtPCA	MidpointKMeans	120	3.5	72.247	-0.091	0.083	1.796	-0.220	0.250	1.808
fGGtPCA	NodePair	—	3.5	70.856	-0.095	0.167	2.483	-0.248	0.167	1.342
fGGtPCA	MidpointKMeans	120	4.5	70.464	-0.082	0.125	0.752	-0.207	0.208	1.221
fGGtPCA	MidpointKMeans	80	3.5	68.288	-0.026	0.000	1.292	-0.222	0.125	1.382
fGGtPCA	NodePair	—	4.5	65.892	-0.086	0.083	0.491	-0.179	0.167	0.686
fGGtPCA	MidpointKMeans	120	5.5	65.655	-0.040	0.042	0.560	-0.200	0.125	1.252
fGGtPCA	MidpointKMeans	80	5.5	65.594	-0.007	0.000	0.047	-0.219	0.167	1.162
fGGtPCA	MidpointKMeans	40	3.5	65.477	-0.073	0.042	0.237	-0.271	0.250	0.690
fGGtPCA	MidpointKMeans	80	4.5	64.833	-0.011	0.000	1.108	-0.183	0.083	0.428
fGGtPCA	NodePair	—	5.5	59.918	-0.033	0.042	0.034	-0.149	0.083	0.255
fGGtPCA	MidpointKMeans	40	4.5	59.594	-0.040	0.042	0.000	-0.198	0.167	0.169
fGGtPCA	MidpointKMeans	40	5.5	53.930	-0.049	0.042	0.024	-0.211	0.167	0.318
GGtPCA	MidpointKMeans	120	3.5	24.857	-0.019	0.000	0.255	-0.065	0.042	0.112
GGtPCA	MidpointKMeans	120	4.5	24.296	-0.070	0.083	0.000	-0.088	0.042	0.000
GGtPCA	MidpointKMeans	80	4.5	24.295	-0.010	0.000	0.000	-0.062	0.000	0.000
GGtPCA	MidpointKMeans	80	3.5	22.774	-0.021	0.000	0.159	-0.100	0.042	0.001
GGtPCA	MidpointKMeans	80	5.5	21.973	-0.006	0.000	0.000	-0.062	0.042	0.000
GGtPCA	NodePair	—	4.5	20.567	-0.077	0.125	1.232	-0.108	0.042	0.895
GGtPCA	MidpointKMeans	120	5.5	18.145	-0.040	0.042	0.000	-0.088	0.042	0.000
GGtPCA	NodePair	—	5.5	17.800	-0.034	0.083	0.179	-0.041	0.042	0.014
GGtPCA	MidpointKMeans	40	3.5	16.586	-0.087	0.000	0.182	-0.106	0.083	0.043
GGtPCA	NodePair	—	3.5	15.330	-0.118	0.167	6.919	-0.123	0.125	1.099
GGtPCA	MidpointKMeans	40	4.5	11.550	-0.071	0.042	0.000	-0.097	0.083	0.000
GGtPCA	MidpointKMeans	40	5.5	3.354	-0.062	0.042	0.000	-0.083	0.042	0.000

Table 4: Sensitivity-analysis results for all tested GGtPCA and fGGtPCA configurations. Primary selection metrics are grouped separately from secondary supporting metrics. The configurations selected for the final baseline comparison are shown in bold.

Pipeline	Artefact Suppression			Signal Preservation				Over-correction			
	SHM att. (%)	SHM $\Delta\rho_{task}$	SHM Δf_{task}^-	NHM $\Delta\rho_{full}$	NHM Δf_{full}^-	SHM $\Delta\rho_{rest}$	SHM Δf_{rest}^-	NHM HbO Rest (%)	NHM HbR Rest (%)	SHM HbO Rest (%)	SHM HbR Rest (%)
No correction	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
TDDR	90.895	-0.228	0.167	0.252	-0.208	0.258	-0.250	-15.689	-5.272	-17.112	-38.614
GGtPCA	24.296	-0.088	0.042	-0.070	0.083	-0.064	0.042	0.000	0.000	0.000	0.000
fGGtPCA	65.594	-0.219	0.167	-0.007	0.000	-0.078	0.000	-0.013	0.081	-1.644	-0.679

Table 5: Pipeline comparison across three evaluation dimensions. Artefact suppression measures motion-period attenuation and task-segment correlation change. Signal preservation uses NHM full-recording and SHM rest-period correlation metrics. Over-correction is assessed via HbO and HbR rest-period signal standard-deviation median change in both NHM and SHM conditions.

Appendix

A Correlation Coefficient Threshold Derivation

The MBLL inversion produces HbO and HbR as linear combinations of the two optical density signals:

$$\Delta\text{HbO} = a \cdot \Delta\text{OD}_{\lambda_1} + b \cdot \Delta\text{OD}_{\lambda_2} \quad (1)$$

$$\Delta\text{HbR} = c \cdot \Delta\text{OD}_{\lambda_1} + d \cdot \Delta\text{OD}_{\lambda_2} \quad (2)$$

where a, b, c, d are the elements of the inverted extinction coefficient matrix scaled by the differential pathlength factor and source-detector distance. The covariance of the two outputs is

$$\text{Cov}(\Delta\text{HbO}, \Delta\text{HbR}) = ac\sigma_1^2 + bd\sigma_2^2 + (ad + bc)\text{Cov}(\Delta\text{OD}_{\lambda_1}, \Delta\text{OD}_{\lambda_2}) \quad (3)$$

Assuming equal variance across wavelengths ($\sigma_1^2 = \sigma_2^2 = \sigma^2$) and writing $\text{Cov}(\Delta\text{OD}_{\lambda_1}, \Delta\text{OD}_{\lambda_2}) = r\sigma^2$ where r is the Pearson correlation coefficient between the two OD signals, this simplifies to

$$\text{Cov}(\Delta\text{HbO}, \Delta\text{HbR}) = [(ac + bd) + (ad + bc)r]\sigma^2 \quad (4)$$

Setting $\text{Cov}(\Delta\text{HbO}, \Delta\text{HbR}) = 0$ and solving for r gives the crossover threshold above which HbO and HbR become positively correlated:

$$r^* = -\frac{ac + bd}{ad + bc} \quad (5)$$

For the extinction coefficients at 735 nm and 850 nm used in this study (from Prahl's tabulated values, scaled by $\ln(10)/10$ for unit conversion), the inverse matrix yields $a = -0.00341$, $b = 0.00522$, $c = 0.00544$, $d = -0.00204$, giving $r^* \approx 0.83$. When the inter-wavelength OD correlation in the haemodynamic band exceeds this value, the MBLL inversion can no longer meaningfully separate the two chromophores and instead produces positively correlated outputs. This threshold is specific to the wavelength pair used and would differ for other instrument configurations.

References

- [1] Arefeh Sherafati, Abraham Z Snyder, Adam T Eggebrecht, Karla M Bergonzi, Tracy M Burns-Yocum, Heather M Lugar, Silvina L Ferradal, Amy Robichaux-Viehoever, Christopher D Smyser, Ben J Palanca, et al. Global motion detection and censoring in high-density diffuse optical tomography. *Human brain mapping*, 41(14):4093–4112, 2020.
- [2] Meryem A Yücel, Juliette Selb, Robert J Cooper, and David A Boas. Targeted principle component analysis: a new motion artifact correction approach for near-infrared spectroscopy. *Journal of innovative optical health sciences*, 7(02):1350066, 2014.
- [3] Frank A Fishburn, Ruth S Ludlum, Chandan J Vaidya, and Andrei V Medvedev. Temporal derivative distribution repair (tddr): a motion correction method for fnirs. *Neuroimage*, 184:171–179, 2019.
- [4] Sergio Fantini, Dennis Hueber, Maria Angela Franceschini, Enrico Gratton, Warren Rosenfeld, Phillip G Stubblefield, Dev Maulik, and Miljan R Stankovic. Non-invasive optical monitoring of the newborn piglet brain using continuous-wave and frequency-domain spectroscopy. *Physics in Medicine & Biology*, 44(6):1543–1563, 1999.
- [5] Meltem Izzetoglu, Prabhakar Chitrapu, Scott Bunce, and Banu Onaral. Motion artifact cancellation in nir spectroscopy using discrete kalman filtering. *Biomedical engineering online*, 9(1):16, 2010.
- [6] Ruisen Huang, Keum-Shik Hong, Dalin Yang, and Guanghao Huang. Motion artifacts removal and evaluation techniques for functional near-infrared spectroscopy signals: a review. *Frontiers in Neuroscience*, 16:878750, 2022.
- [7] Hubin Zhao, Elisabetta M Frijia, Ernesto Vidal Rosas, Liam Collins-Jones, Greg Smith, Reuben Nixon-Hill, Samuel Powell, Nicholas L Everdell, and Robert J Cooper. Design and validation of a mechanically flexible and ultra-lightweight high-density diffuse optical tomography system for functional neuroimaging of newborns. *Neurophotonics*, 8(1):015011–015011, 2021.
- [8] Xu Cui, Signe Bray, and Allan L Reiss. Functional near infrared spectroscopy (nirs) signal improvement based on negative correlation between oxygenated and deoxygenated hemoglobin dynamics. *Neuroimage*, 49(4):3039–3046, 2010.
- [9] Sabrina Brigadoi, Lisa Ceccherini, Simone Cutini, Fabio Scarpa, Pietro Scatturin, Juliette Selb, Louis Gagnon, David A Boas, and Robert J Cooper. Motion artifacts in functional near-infrared spectroscopy: a comparison of motion correction techniques applied to real cognitive data. *Neuroimage*, 85:181–191, 2014.
- [10] Felix Scholkmann, Sonja Spichtig, Thomas Muehlemann, and Martin Wolf. How to detect and reduce movement artifacts in near-infrared imaging using moving standard deviation and spline interpolation. *Physiological measurement*, 31(5):649–662, 2010.
- [11] Behnam Molavi and Guy A Dumont. Wavelet-based motion artifact removal for functional near-infrared spectroscopy. *Physiological measurement*, 33(2):259–270, 2012.
- [12] Yiheng Zhang, Dana H Brooks, Maria Angela Franceschini, and David A Boas. Eigenvector-based spatial filtering for reduction of physiological interference in diffuse optical imaging. *Journal of biomedical optics*, 10(1):011014–011014, 2005.
- [13] Costanza Iester, Laura Bonzano, Monica Biggio, Simone Cutini, Marco Bove, and Sabrina Brigadoi. Comparing different motion correction approaches for resting-state functional connectivity analysis with functional near-infrared spectroscopy data. *Neurophotonics*, 11(4):045001–045001, 2024.
- [14] Fan Zhang, Adaira Reid, Alissa Schroeder, Lei Ding, and Han Yuan. Controlling jaw-related motion artifacts in functional near-infrared spectroscopy. *Journal of Neuroscience Methods*, 388:109810, 2023. ISSN 0165-0270. doi: <https://doi.org/10.1016/j.jneumeth.2023.109810>. URL <https://www.sciencedirect.com/science/article/pii/S0165027023000298>.
- [15] Yunjia Xia, Jianan Chen, Jinchen Li, Tingchen Gong, Ernesto E Vidal-Rosas, Rui Loureiro, Robert J Cooper, and Hubin Zhao. A deep-learning empowered, real-time processing platform of fnirs/dot for brain computer interfaces and neurofeedback. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2025.
- [16] Paola Pinti, Ilias Tachtsidis, Antonia Hamilton, Joy Hirsch, Clarisse Aichelburg, Sam Gilbert, and Paul W Burgess. The present and future use of functional near-infrared spectroscopy (fnirs) for cognitive neuroscience. *Annals of the new York Academy of Sciences*, 1464(1):5–29, 2020.
- [17] Hayder R Al-Omairi, Sebastian Fudickar, Andreas Hein, and Jochem W Rieger. Improved motion artifact correction in fnirs data by combining wavelet and correlation-based signal improvement. *Sensors*, 23(8):3979, 2023.

- [18] Hanfei Li, Chenyu Fan, Ke Chen, Hongyu Xie, Guohui Yang, Haozheng Li, Xiangtong Ji, Yi Wu, and Meng Li. Brain activation during motor preparation and execution in patients with mild cognitive impairment: An fnirs study. *Brain Sciences*, 15(4):333, 2025.
- [19] Paola Pinti, Felix Scholkmann, Antonia Hamilton, Paul Burgess, and Ilias Tachtsidis. Current status and issues regarding pre-processing of fnirs neuroimaging data: an investigation of diverse signal filtering methods within a general linear model framework. *Frontiers in human neuroscience*, 12:505, 2019.
- [20] Sabrina Brigadoi and Robert J Cooper. How short is short? optimum source–detector distance for short-separation channels in functional near-infrared spectroscopy. *Neurophotonics*, 2(2):025005–025005, 2015.
- [21] Luca Pollonini, Heather Bortfeld, and John S Oghalai. Phoebe: a method for real time mapping of optodes-scalp coupling in functional near-infrared spectroscopy. *Biomedical optics express*, 7(12):5104–5119, 2016.
- [22] Franziska Klein. Optimizing spatial specificity and signal quality in fnirs: an overview of potential challenges and possible options for improving the reliability of real-time applications. *Frontiers in Neuroergonomics*, 5:1286586, 2024.
- [23] Xin Zhou, Gabriel Sobczak, Colette M McKay, and Ruth Y Litovsky. Comparing fnirs signal qualities between approaches with and without short channels. *PloS one*, 15(12):e0244186, 2020.
- [24] Lia M Hocke, Ibukunoluwa K Oni, Chris C Duszynski, Alex V Corrigan, Blaise deB Frederick, and Jeff F Dunn. Automated processing of fnirs data—a visual guide to the pitfalls and consequences. *Algorithms*, 11(5):67, 2018.
- [25] Shijing Wu, Jun Li, Lantian Gao, Changshui Chen, and Sailing He. Suppressing systemic interference in fnirs monitoring of the hemodynamic cortical response to motor execution and imagery. *Frontiers in Human Neuroscience*, 12:85, 2018.
- [26] Scott Prahl. Tabulated molar extinction coefficient for hemoglobin in water. OMLC. Accessed 2026-03-25.
- [27] Georgina Leadley, Robert J Cooper, Topun Austin, Jeremy C Hebden, and Gemma Bale. Investigating the effect of limited spectral information on nirs-derived changes in hemoglobin and cytochrome-c-oxidase concentration with a diffusion-based model. *Biomedical Optics Express*, 15(10):5912–5931, 2024.
- [28] MNE-Python developers. mne.preprocessing.nirs.beer_lambert_law. MNE-NIRS documentation / source code. Accessed 2026-03-25.